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Junior, familyi=P. J., given=José Eduardo, giveni=J. E., givenun=0un=0,uniquepart=base,hash=f8590
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familyi=S., given=Leena, giveni=L., givenun=0 au-
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Algoritmos de optimización de problemas de Producción Industrial y Logística en la Industria 4.0

Autor: Juan Moreno Malo

Director: D. Josefa Mula Bru
D. Raul Poler

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Chapter 1

Route Clustering Algorithms for Procurement Transport

1.1 Introduction

Maximising economic potential and social benefits, and minimising environmental impacts are critical objectives in the efficient utilisation of productive and logistics resources in manufacturing supply chains. Procurement, as defined by **Chopra2015SupplyOperation**, is the process by which goods and services are acquired in a supply chain, whose main aim is to source raw materials, components, or finished products for a factory or distribution centre from suppliers. Note that while the supplier "typically" assumes the responsibility for transport planning, there are cases, such as the automotive industry, in which manufacturers actively participate in making transport decisions due to the complex interactions between numerous manufacturing facilities and suppliers **Fleischmann2015TransportDistribution**. This indicates that transport decisions during procurement should be a collaborative effort made among all supply chain parties to ensure that the sourcing process is efficiently managed. Here we focus on optimising the tactical decisions involved in procurement transport planning by underscoring the importance of strategic collaboration by aggregating the transport resources of different routes to enhance supply chain efficiency.

According to **Diaz-Madronero2014AChain**, automatic algorithm-based procurement transport planning is preferable to manual planning based on planners' judgement and experience. The latter can be myopic and generate suboptimal plans. Such insight should be amplified when considering the optimi-

sation of procurement transport problems when there are many nodes in the supply chain, where the coordination of the different stakeholders can reduce logistics, transportation costs and the environmental impact. One possible way to coordinate these logistic processes is to cluster the routes between suppliers and manufacturing facilities so that the routes in the same cluster share transportation resources. This is of outmost importance when routes are long and constrained by weight and volume because we can mix the required goods inside trucks to minimise the number of trucks. Hence this article aims to bridge this research gap by providing a novel mixed integer linear mathematical programming (MILP) model that clusters the routes between the suppliers and manufacturing facilities to minimise the procurement transport cost along the supply chain. Additionally, as a solution approach, branch and price algorithms is developed. Branch and price combine two methods: branch-and-bound and column generation (CG). It is often applied to problems like vehicle routing, crew scheduling, and cutting stock problems. CG is a technique to solve large MILP models with a determined decomposable structure. It was introduced in **Ford1958ConstructingFlows** to address a multi-commodity network flow problem. Then, **Dantzig1960DecompositionPrograms** adapted it to LP problems with a decomposable structure. Furthermore, **Gilmore1961AProblem** demonstrated the efficiency of the CG technique to the cutting stock problem. For a detailed explanation about the inner workings of CG and branch and price techniques we refer readers to **Lubbecke2005SelectedGeneration** and **Desrosiers2011BranchPriceandCutAlgorithms**. Here, a solution approach based on a branch and price algorithm using column generation is developed to solve real-world instances with up to 900 suppliers, 50 manufacturing facilities and 1,800 routes.

The main contributions of this paper are: (i) formulating a novel MILP model for the procurement transport supply chain route clustering, dubbed as SCRC, that integrates volume and weight distribution; (ii) developing a branch and price algorithm to find good quality solutions in a reasonable amount of time; (iii) validating and testing the SCRC model with a case study at a first-tier supplier of the automotive industrial sector. Readers are also referred to the literature reviews of **Mosca2019IntegratedReview** and **Diaz-Madronero2015ADecisions**, which focus on the integration of inventory decisions and transport decisions.

The rest of the paper is organised as follows. Section 2 presents the literature review. Section 3 describes the addressed problem. Section 4 introduces the mathematical formulation of the SCRC. Section 5 describes the branch and price procedure. Section 6 includes the computational experiments. Section 7 explains how the SCRC can be used as a decision support tool. Section 8 concludes the research and analyses future research lines.

1.2 Literature Review

Transport processes are, extensively, explored in academic research due to their critical role in supply chains. As highlighted by **Langley2020SupplyPerspective**, transportation is essential for transferring raw materials and final products across different supply chain stages, including suppliers, warehouses, manufacturers, and retailers. **Fleischmann2015TransportDistribution** categorise transport processes in supply chains into two main types: the transport of materials from suppliers to manufacturing facilities, known as procurement transport; the distribution of finished goods from factories to consumers. This discussion focuses on addressing issues related to procurement transport.

Procurement transport problems can be analysed with two different dimensions: the configuration of the supply chain and the used transportation method. When considering supply chain configurations, **Mosca2019IntegratedReview** distinguish among four different scenarios: (i) from a single source to a single destination (SS); (ii) from a single source to multiple destinations (SM); (iii) from multiple sources to a single destination (MS); (iv) from multiple sources to multiple destinations (MM). The "source" reflects the number of considered supplier locations, whereas the "destination" indicates the number of manufacturing sites receiving supplies. Additionally, **Diaz-Madronero2014AChain** highlight various transportation methods, including full-truckload (FTL), less-than-truckload (LTL) and milk run. Here the aggregation of transportation resources in the scientific literature is also a main concern.

Schoneberg2010AnNetworks study the concept of delivery profiles, which determine the frequency at which deliveries are made and are set at a fixed rate. In this context, FTL transport occurs when the volume of goods exceeds the capacity of one vehicle and, therefore, direct shipment is a viable option. LTL is used when the volume of goods does not fill an entire truck. In this case, goods are shipped to a consolidation centre before being transported to their final destination. The proposed model is designed for the MS supply chain configuration and routes are consolidated by LTL. In **Schoneberg2013ANetworks**, the proposed model considers demand uncertainty.

Diaz-Madronero2017ADecisions introduce an MILP model to integrate production, procurement and transport planning decisions into manufacturing systems. The proposed model goes beyond traditional material requirement planning systems by considering the MS supply chain configuration in a unified optimisation framework. It takes into account the capacities of material and production resources, as well as procurement transport planning decisions. Routes can be consolidated using three modes of transport: FTL, LTL and milk run.

Chitsaz2019ARouting address integrating production, inventory and procurement transport into a manufacturing context by defining this problem as an assembly routing problem (ARP). It introduces an MILP model for the ARP

and proposes a three-phase decomposition matheuristic as a solution approach. Routes are consolidated using a milk-run scheme to procure products.

Baller2022OptimizingApproach study the optimisation of inbound transportation of an automotive factory from first-tier suppliers. The MS study uses FTL, LTL and courier express service (CES) as transport methods. Routes are consolidated using LTL as a transport method.

Bou-Zeid2023OptimalIndustry. examine a multiperiod tactical transportation plan by focusing on the inbound logistics for an automobile manufacturer. The study explores direct, indirect, and milk-run transportation strategies, each characterised by its unique cost structure. The objective is to determine the most efficient routes and strategies to optimise the transportation flow. The concept of clustering routes is introduced into an MS supply chain configuration.

Table **??** compares the supply chain configuration, transport modelling, and transport aggregation between the SCRC model proposal and those currently in the literature.

Table 1.1: Comparison of procurement transport optimisation models

Article	SC configuration	Decision level	Transportation mode	Transportation capacity constraint	Volume and weight optimisation	Source aggregation	Destination aggregation	Modelling framework
Schoneberg2010AnNetworks	MS	T	FTL|LTL	V		X		MILP
Schoneberg2013ANetworks	MS	O	FTL|LTL	V		X		MILP
Diaz-Madronero2017ADecisions	MS	T	FTL|LTL|MR	V		X		MILP
Chitsaz2019ARouting	MS	O|T	MR	V		X		MILP
Baller2022OptimizingApproach	MS	O	MR|FTL|LTL|CES	V|W	X	X		MILP
Bou-Zeid2023OptimalIndustry.	MR|FTL|LTL	T	MR	V				DA
SCRC	MM	T	FTL|LTL	V|W	X	X	X	MILP

Abbreviations. T (tactical), **O** Operational; **FTL** (full-truck load), **LTL** (less-than-truck load), **MR** (milk-run), **CES** (courier express service); **V** (volumetric), **W** (weight) **UH** Unit Holding; **DA** Decomposition Approach;

The main novelty of our proposal is the joint transport MM optimisation, i.e., between multiple sources and multiple destinations, while optimising the volume and weight mix using the FTL and LTL transport methods.

1.3 Problem Description

The objective of the proposed model is to reduce transportation costs in an MM supply chain by consolidating transport from different suppliers to various manufacturing plants from a tactical decision level. Figure **??** illustrates the considered supply chain, where circles are supply chain nodes and edges represent the established routes between them. Therefore, a node can be both a manufacturing facility and a supplier for other nodes, depending if the node has incoming and outcoming edges. Each route has a preset transport type, number of used trucks and volume and weight transported per week to procure the different manufacturing facilities. Depending on these parameters and route length, there is also a pre-established cost per week.

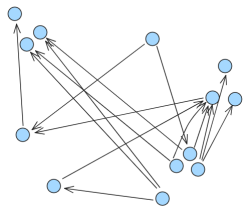


Figure 1.1: Routes of the supply chain

If a route is in low demand, i.e., the number of trucks needed to transport goods per week is less than one, then the route is normally delegated to a logistics company with higher shipping costs. Otherwise, the employed transport type is FTL. Efficiency rarely reaches 100% due to varying demands or the need for urgent sourcing, which necessitate more trucks or sending partially filled trucks. Moreover, the minimum required number of trucks is determined by the route's total volume and weight, and the truck's maximum capacity in both volume and weight terms. Jointly managing the clustered routes offers the possibility of reducing the current costs. On the one hand, we eliminate the cost of delegating the routes in low demand because we consolidate all the routes in the cluster. On the other hand, we can reduce the number of trucks needed by mixing the routes dominated by weight and volume.

To optimise transportation decisions, we propose the concept of clusters by grouping routes for joint transportation management purposes. A cluster comprises both the receiving and sourcing nodes. We set both the receiving and sourcing central hubs nodes of the supply chain. The transport in the cluster can be separated into three stages: sourcing to the source central hub; transporting goods from the source central hub to the receiving central hub; distributing goods from the receiving central hub to the receiving nodes. The transportation cost associated with this cluster is derived in these three stages. Figure **??** illustrates the clustering process of a set of routes.

Additionally, each route has a set of routes to which it can be clustered. This is because some routes carry materials that are incompatible for joint transport. Furthermore, the beginning or end of routes could be in different countries, and routes are not mixed because of customs.

The objective of the proposed optimisation method is to find new clusters to minimise the transportation costs. Not all routes need to be clustered. Therefore, a solution is determined by the sets of clusters and non-clustered routes. A possible solution for the supply chain example is illustrated in Figure **??**. Note that not all routes have been clustered.

(a) Routes **(b)** Cluster

Figure 1.2: Clustered routes example

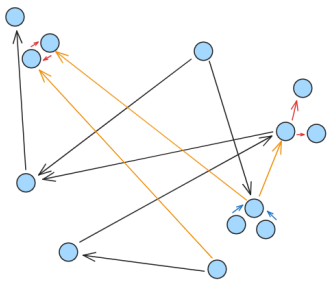


Figure 1.3: Clustered supply chain

The problem can be stated as follows:
Initial information:

- Transport details of routes (weight, volume, current cost).
- Distance among all the nodes of the supply chain.
- Set of edges that can be grouped together.
- Cost per kilometre.
- Maximum weight and volume allowed per truck.
- Parameters concerning consolidation efficiency.

To determine:

- Set of clusters for transport consolidation.
- Set of unaltered routes.

With constraints:

- Adhering to the association graph.

Objective:

- Minimise transportation costs.

1.4 Mathematical Formulation

In this section, we describe the developed SCRC model. The nomenclature is provided in Table **??**

Table 1.2: Nomenclature of SCRC

Indices	
V	Set of vertices of the graph (v =1, 2,. . . ,V)
E	Set of edges of the graphs that represent the current routes (e =1, 2,. . . ,E)
C	Set of edge cluster (e =1, 2,. . . ,C)
Set of Indices	
$\Gamma(e)$	Set of edges that can be grouped with e
$\delta^+(e)$	Receiving node of the edge
$\delta^-(e)$	Source node of the edge
Parameters	

v_e	Volume transported on route e
w_e	Weight transported on route e
d_{ij}	Distance between nodes $i \in V$ and $j \in V$
c_e	Initial cost of route on edge e
max_v	Maximum volume carried by a truck
max_w	Maximum weight carried by a truck
α	Efficiency of transport where volume dominates the route
β	Efficiency of transport where weight dominates the route
γ	Efficiency of transport between hubs
γ^+	Efficiency of transport between receiving hubs and receiving nodes
γ^-	Efficiency of transport between sourcing nodes and sourcing hub
Decision Variables	
Y_e	Binary variable indicating if e is not clustered
X^c	Binary variable indicating if cluster c is used
X_{ce}	Binary variable indicating if edge e belongs to cluster c
X^-_{ce}	Binary variable indicating if node $\delta^-(e)$ is the sourcing hub for the sourcing nodes in cluster c
X^+_{ce}	Binary variable indicating if node $\delta^+(e)$ is the receiving hub for the receiving nodes
$X^e_{ce'e''}$	Binary variable indicating if the edge e is in c and the source node of e' is the sourcing hub for outgoing goods and receiving node of e'' if the is the receiving hub for incoming goods
D_{ce}	Distance between the sourcing and receiving hub of cluster c if e is in the cluster, 0 if not.
D^-_{ce}	Distance from $\delta^-(e)$ to the sourcing hub if e is in cluster c
D^+_{ce}	Distance from $\delta^+(e)$ to the receiving hub if e is in cluster c
K_c	Transportation cost between sourcing hub and receiving hub at cluster c
K^-_c	Transportation cost between sourcing nodes and sourcing hub at cluster c
K^+_c	Transportation cost between receiving hub and receiving nodes at cluster c

The objective is to minimise transportation costs, i.e., the cost of the non-altered routes and the cost of the different clusters.

$$\min z = \sum_{e \in E} Y_e c_e + \sum_c \gamma K_c + \gamma^+ K_c^- + \gamma^- K_c \tag{1.1}$$

subject to:

$$Y_e + \sum_c X_{ce} = 1 \quad \forall e \in E \tag{1.2}$$

$$\sum_{e \in E} X_{ce} \leq M X^c \quad \forall c \in C \tag{1.3}$$

$$\sum_{e \in E} X_{ce} \geq 2X^c \quad \forall c \in C \tag{1.4}$$

$$\sum_{e \in E} X^-_{ce} = X^c \quad \forall c \in C \tag{1.5}$$

$$\sum_{e \in E} X^+_{ce} = X^c \quad \forall c \in C \tag{1.6}$$

$$\sum_{e' \in \Gamma(e)} X^+_{ce'} = X_{ce} \quad \forall c \in C \tag{1.7}$$

$$\sum_{e' \in \Gamma(e)} X^-_{ce'} = X_{ce} \quad \forall c \in C \tag{1.8}$$

$$\sum_{e', e'' \in \Gamma(e)} X^e_{ce'e''} = X_{ce} \quad \forall c \in C, e \in E \tag{1.9}$$

$$X^e_{ce'e''} + 2(1 - X_{ce}) + 1 \geq X^-_{ce'} + X^+_{ce''} \quad \forall c \in C, e \in E, e', e'' \in \Gamma(e) \tag{1.10}$$

$$\sum_{e' \in (E'\Gamma(e))} X_{ce'} \leq M(1 - X_{ce}) \quad \forall c \in C, e \in E \tag{1.11}$$

$$D^-_{ce} = \sum_{e' \in \Gamma(e)} X^-_{ce'} d_{\delta^-(e)\delta^-(e')} \quad \forall c \in C, e \in E \tag{1.12}$$

$$D^+_{ce} = \sum_{e' \in \Gamma(e)} X^-_{ce'} d_{\delta^+(e)\delta^+(e')} \quad \forall c \in C, e \in E \tag{1.13}$$

$$D_{ce} = \sum_{e', e'' \in \Gamma(e)} X^e_{ce'e''} d_{\delta^-(e')\delta^+(e'')} \quad \forall c \in C, e \in E \tag{1.14}$$

$$max_v K^-_{ce} \geq D^-_{ce} v_e \quad \forall c \in C, e \in E \tag{1.15}$$

$$max_w K^-_{ce} \geq D^-_{ce} w_e \quad \forall c \in C, e \in E \tag{1.16}$$

$$K^-_c = \sum_{e \in E} K^-_{ce} \quad \forall c \in C \tag{1.17}$$

$$max_v K^+_{ce} \geq D^+_{ce} v_e \quad \forall c \in C, e \in E \tag{1.18}$$

$$max_v K^+_{ce} \geq D^+_{ce} v_e \quad \forall c \in C, e \in E \tag{1.19}$$

$$K^+_c = \sum_{e \in E} K^+_{ce} \quad \forall c \in C \tag{1.20}$$

$$max_w K_c \geq (\sum_{e \in E} D_{ce} w_e)\alpha \quad \forall c \in C \tag{1.21}$$

$$max_v K_c \geq (\sum_{e \in E} D_{ce} v_e)\beta \quad \forall c \in C \tag{1.22}$$

Equation (**??**) determines if a route is clustered or not. If a cluster is formed, then it should have at least two routes (Equations (**??**-**??**)). For each created cluster, we should set a sourcing hub, and a receiving hub (Equations (**??**-**??**)). Equation (**??**-**??**) links the sourcing hub and the receiving hub for all the routes in the cluster. Equations (**??**) guarantees that only compatible edges can be set in a cluster. For each clustered route, Equation (**??**) sets the distance from the outgoing node of the route to the sourcing hub. Equation (**??**) sets the distance from the sourcing hub to the receiving hub. Equation (**??**) establishes the distance from the receiving hub to the destination node of the route. Equations (**??**-**??**) determine the transportation cost from the outgoing node to the sourcing hub by taking into account the distance between them and the number of trucks needed, which depends on the volume and weight of the routes. Equation (**??**) determines the total transportation costs to the supplier hub. Similarly, Equations (**??**-**??**) determine the transportation costs of the cluster from the receiving hub to the destination nodes. Equations (**??**-**??**) determine the transportation cost from the sourcing hub to the receiving hub by taking into account the distance between them and the number of trucks needed, which depends on the volume and weight of routes and the efficiency parameters.

1.5 Branch and Price Solution Approach

There are two main resolution issues with the provided SCRC model. The first one is that a priori we do not know the exact number of needed clusters. Therefore, we could overestimate the number of clusters needed to find the optimal clustering of routes. The other is that the number of needed variables and constraints grows linearly in relation to the number of clusters set. Meanwhile, the decision space of the integer variables exponentially grows. This leads to an inefficient solution using current MILP solvers. Therefore, a column generation scheme based on branch and price is developed.

The key components of the branch and price algorithm are the master problem (MP), the restricted master problem (RMP), the pricing problem (PP) and the branching policy. The MP is a reformulation of a linear or integer programming model so that we can exploit the decomposable structure of the problem. However, the MP is typically too large to be solved directly because it involves a huge number of variables and columns that represent all the possible decision scenarios (e.g., all the possible routes in a routing problem or all the feasible clusters in our case). To manage this complexity, we consider only a smaller subset of variables, which leads to the formulation of the RMP to find new promising columns to be added to the RMP. The RMP only uses a subset of the variables and columns of the MP. Solving the RMP provides an optimum solution for the included subset of variables, but not necessarily for the entire set of original variables. Therefore, RMP integrality constraints are relaxed to find the current dual variables and new columns through the PP, also known as the subproblem, which is used to identify

which variables (or columns) should be added to the RMP to potentially improve the current solution. The goal of the PP is to find variables with a negative reduced cost (in a minimisation problem) to be added to the RMP, which is likely to lead to a better objective value.

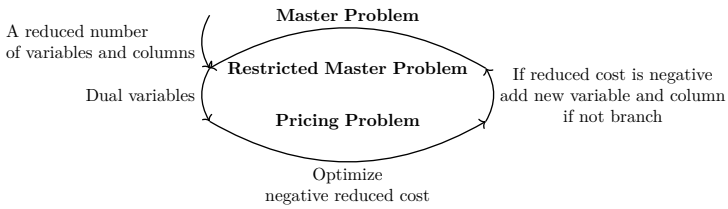


Figure 1.4: Branch and Price

If there are no new columns with a negative reduced cost, then we have optimally solved the integrality relaxed MP. If all the variables associated with the solution are integers, then we have found an integer solution to the current branch. If not, we have to branch the current branch by adding further constraints to find integer solutions through the relaxed RMP. Figure **??** exmplifies the Branch & Price Algorithm.

1.5.1 Formulation of the Master Problem and the Reduced Master Problem

Here we firstly reformulate the model as a set-partitioning formulation (Dantzig – Wolfe decomposition). The key idea of the set-partitioning formulation for the SCRC is to enumerate all the feasible clusters with their associated costs. Let P be the set of all feasible clusters that satisfy the constraints imposed by the SCRC model. For each cluster $p \in P$, a binary variable λ_p is defined that equals to 1 if cluster p appears in the solution and 0 otherwise. Binary parameter a_e^p equals to 1 if cluster p contains the edge $e \in E$. Parameter c_p represents the cost associated with cluster c. In this case, we consider that the pre-established routes are clusters with pre-established cost and with $a_e^p = 1$ if e is the current route, and 0 otherwise. Therefore, the MP of our problem is:

$$\min \sum_{p \in P} c_p \lambda_p \tag{1.23}$$

subject to:

$$\sum a_e^p \lambda^p = 1 \quad \forall e \in E \tag{1.24}$$

$$\lambda_p \in \{0, 1\} \quad \forall p \in P \tag{1.25}$$

The objective function **??** is to minimise the cost of the clusters needed to transport goods. Meanwhile, **??** ensures that each route is only used in one cluster.

Here we employ the initial routes as the initial clusters used in the RMP with their associated costs.

$$\min \sum_{p \in P} c_p \lambda_p \tag{1.26}$$

subject to:

$$\sum a_e^p \lambda^p = 1 \quad \forall e \in E \tag{1.27}$$

$$0 \leq \lambda_p \leq 1 \quad \forall p \in P \tag{1.28}$$

Dual values are obtained by solving the RMP. Let $\pi_e, e \in E$, be dual variables associated with constraints (**??**). Then the reduced cost associated with cluster $p \in P$ is $c_p - \sum_e a_e^p \pi_e$. The feasible cluster with a negative reduced cost can improve the current solution to the RMP. For each iteration, we solve the subproblem that consists of finding a feasible cluster with the minimum negative reduced cost. The value of the variables associated with negative reduced cost (expressed as new columns) are added to the RMP. Then the RMP is solved to generate a new set of dual values. This process is repeated until no further variables with a negative reduced cost are discovered.

1.5.2 Pricing Subproblem

We solve the pricing problem using an MILP approach. The indices and parameters are the same as the model proposed before, but the decisions variables do not depend on the indices of clusters. Table **??** describes the new decision variables.

Table 1.3: Nomenclature of the pricing subproblem

Decision Variables	
X_e	Binary variable indicating if edge e belongs to the new cluster
X_e^-	Binary variable indicating if $\delta^-(e)$ is the sourcing hub for the suppliers in the new cluster
X_e^+	Binary variable indicating if $\delta^+(e)$ is the receiving hub for the nodes in the new cluster
$X_{e'e''}^e$	Binary variable indicating if edge e is in c and the source node of e' is the sourcing hub for outgoing goods and receiving node of e'' is receiving hub for incoming goods in the new cluster
D_e	Distance between the sourcing and receiving hub of the new cluster if e is in the cluster, 0 if not.
D_e^-	Distance from $\delta^-(e)$ to the sourcing hub if e is in the new cluster, 0 if not
D_e^+	Distance from $\delta^+(e)$ to the receiving hub if e is in the new cluster, 0 if not
K	Transportation cost between the sourcing nodes and the sourcing hub in the new cluster

K^-	Transportation cost between the sourcing hub and the receiving hub in the new cluster
K^+	Transportation cost between the receiving hub in the new cluster

The objective is to find new clusters with negative reduced costs. That is:

$$\min z = \gamma K + \gamma^- K^- + \gamma^+ K^+ - \sum_{e \in E} \pi_e * X_e \tag{1.29}$$

subject to:

$$\sum_{e \in E} X_e \geq 2 \tag{1.30}$$

$$\sum_{e \in E} X_e^- = 1 \tag{1.31}$$

$$\sum_{e \in E} X_e^+ = 1 \tag{1.32}$$

$$\sum_{e' \in \Gamma(e)} X_{e'}^+ = X_e \quad \forall e \in E \tag{1.33}$$

$$\sum_{e' \in \Gamma(e)} X_{e'}^- = X_e \quad \forall e \in E \tag{1.34}$$

$$\sum_{e', e'' \in \Gamma(e)} X_{e'e''}^e = X_e \quad \forall e \in E \tag{1.35}$$

$$X_{e'e''}^e + 2(1 - X_e) + 1 \geq X_{e'}^- + X_{e''}^+ \quad \forall e \in E, e', e'' \in \Gamma(e) \tag{1.36}$$

$$\sum_{e' \in E' \Gamma(e)} X_{e'} \leq M(1 - X_e) \quad \forall e \in E \tag{1.37}$$

$$D_e^- = \sum_{e' \in \Gamma(e)} X_{e'}^- d_{\delta^-(e)\delta^-(e')} \quad \forall e \in E \tag{1.38}$$

$$D_e^+ = \sum_{e' \in \Gamma(e)} X_{e'}^- d_{\delta^+(e)\delta^+(e')} \quad \forall e \in E \tag{1.39}$$

$$D_e = \sum_{e', e'' \in \Gamma(e)} X_{ce'e''}^e d_{\delta^-(e')\delta^+(e'')} \quad \forall e \in E \tag{1.40}$$

$$max_v K_e^- \geq D_e^- v_e \quad \forall e \in E \tag{1.41}$$

$$max_w K_e^- \geq D_e^- w_e \quad \forall e \in E \tag{1.42}$$

$$K^- = \sum_{e \in E} K_e^- \tag{1.43}$$

$$max_v K_e^+ \geq D_e^+ v_e \quad \forall e \in E \tag{1.44}$$

$$max_v K_e^+ \geq D_e^+ v_e \quad \forall e \in E \tag{1.45}$$

$$K^+ = \sum_{e \in E} K_e^+ \tag{1.46}$$

$$max_w K \geq (\sum_{e \in E} D_e w_e)\alpha \tag{1.47}$$

$$max_v K \geq (\sum_{e \in E} D_e v_e)\beta \tag{1.48}$$

Similarly, (**??**-**??**) restrict the clusters used in the pricing problem to feasible ones.

Table 1.4: Nomenclature for instance types

Size	Number of Routes
Small	(0, 500]
Medium	(500,1000]
Large	(1000,1500]
Extra Large	(1500-2500]

1.5.3 Branching

Here, we use the depth-first branching rule, which once we can not find any new promising solutions through pricing problem we set the variable with the largest fractional to one. That is, we will fix a cluster that must be used. After fixing the variables, more columns are generated using the pricing algorithm. Repeating this branching strategy leads to an integer solution of the RMP. In our case, this method of branching does not guarantee to find an optimal solution, but it does guarantee a feasible one.

1.6 Computational Experiments

Here we compare both the performance of the SCRC model and the developed price and branch approach. For this purpose, 200 instances of different sizes were artificially generated and inspired in a first-tier supplier of the automotive sector. Depending on the total number of routes, we grouped instances into four main types: small, medium, large and extra larges. The routes in the instances were generated by randomly assigning a source (supplier) and a destination (manufacturing facility) of the supply chain under study. Furthermore, each route was randomly assigned to be constrained by volume or weight.

We used the Julia programming language and the JuMP package to run all the calculations. To solve both the RMP and PP, we have used the Gurobi commercial solver.

The Table **??** illustrates how we have grouped the instances.

Table **??** presents the mean objective value and resolution duration aggregated by the size of the instace for the different resolution procedures. For the SCRC MILP model we have set a time limit of 2 hours. Meanwhile, for the branch and price resolution procudere, we have set a time limit of 30 minutes for each node iteration.

As we can observe, the branch and price algorithm outperforms the MILP model in both terms of solution quality and resolution time.

Table 1.5: Mean objective value and resolution duration per resolution procedure

Type	Resolution pocedure	Transport Cost	Duration
Extra large	B&P	6864750.622806	1826 sec
Extra large	MILP	7507608.574584	11520 sec
Large	B&P	3811390.489451	1803 sec
Large	MILP	4498435.223205	8820 sec
Medium	B&P	2048727.759246	1435 sec
Medium	MILP	2531413.865415	8010 sec
Small	B&P	622577.984265	43 sec
Small	MILP	673251.233579	5825 sec

1.7 SCRC As A Decision Support Tool

The SCRC model can be used as a decision tool for clustering routes of various supply chains. Through the different efficiency parameters of the algorithm, decision makers can analyse the various clustering routes proposed by the algorithm. If we set all the efficiency parameters to 1, then we consider that clustering will not have any inefficiencies, which is rarely the case. Once we have identified where inefficiencies are from, for example, the sourcing to the main hub, then we can adjust this parameter and see how the system behaves. Figure illustrates the different savings proposed by algorithms when we increase the inefficiency of one of the parameters. Furthermore, if a decision maker thinks that a proposed cluster is not well suited for the supply chain being considered, additional constraints can be set in the mathematical model by selecting which routes can a route be clustered.

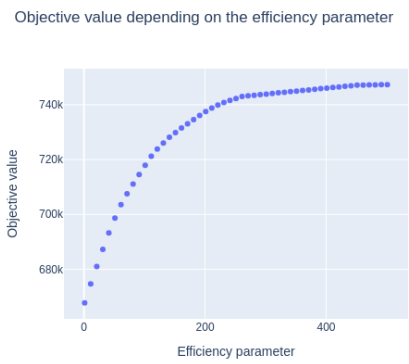


Figure 1.5: Transport cost vs. Efficiency parameter

As we can see in Figure **??**, as the efficiency parameter increases the transport cost also increases. Furthermore, the transport cost can not overpass the initial cost of the problem, since if the new clusters are too costly it will choose to use the initial configuration.

SCRC model has been used as a decision tool for clustering routes at a first tier supplier of the automotive sector.

1.8 Conclusions and Future Research

In conclusion, this paper presents a pioneering effort in route clustering for a supply chain with multiple suppliers and manufacturing facilities. It introduces a mathematical model designed to consolidate the transport resource of large supply chains. Furthermore, the model has different parameters that the decision maker must decide. Not only the development of a mathematical model, but also the subsequent application of a branch and priced method to ascertain near-optimal solutions, underscore the robustness and innovative approach of this study. The decision tool, validated in a first-tier supplier of the automotive industry, provides a solid foundation for future explorations and applications in procurement transport planning.

Looking ahead, future research alternatives have been identified, and promise to further enhance the model's utility and applicability. These include exploring alternative metrics for evaluating the transport cost of each cluster by, for example, adding a vehicle routing problem for sourcing and delivery to the central hubs. Furthermore, it is possible to integrate the model to make operational decisions to dynamically adjust routes based on inventory levels and the demand of the manufacturing facilities. Additionally, it offers the possibility of addressing transport decisions about using reusable packaging in the supply. Regarding the resolution procedure, a future research potential is identified. For instance, using other branching strategies to ensure optimal solutions for medium-sized instances and applying metaheuristcs for the pricing problem for larger instances.

Chapter 2

State of the art Returnable transport items

2.1 Abstract

This study focuses on the use of reusable and returnable containers, also called returnable transport items (RTIs), within the supply chain and the associated transport decisions. Reusable containers function as shared resources among different nodes in the supply chain, enabling efficient storage and flow of raw materials and finished products while reducing environmental impact compared to single-use containers. RTIs offer benefits regarding sustainability, security, and ergonomics for workers. The research analyses how integrating RTI management with transportation decisions can enhance supply chain efficiency, considering other subsystems like production and inventory management. Key research questions include identifying issues in the literature regarding RTIs, evaluating optimisation models for integrating inventory, production, and transportation decisions, and exploring performance indicators in supply chains using RTIs. This research contributes to understanding the impact of reusable containers on supply chain efficiency and provides insights into future research directions.

2.2 Introduction

Supply chain management (SCM) has gained significant attention, with an increasing focus on sustainable practices. One research direction is to optimise the procurement and delivery transport decisions to minimise the carbon footprint and transport cost. Another is to use reusable packaging to reduce the amount of plastic used and the packaging cost. In this study, we focus on the integration of transport decisions and the use of reusable transport items (RTIs) to enhance the efficiency and sustainability of a manufacturing supply chain. RTIs, such as plastic containers, metallic crates, pallets and trays, are an important asset in many companies and industries. Efficient management of RTI closed-loop supply chains not only enables the delivery of finished products and the sourcing of raw materials but also enhances the overall supply chain efficiency, minimising both the economic cost and environmental impact. To our knowledge, only two other systematic literature reviews focused on the quantitative methods relating to the transport decisions of RTIs within supply chains exist (cite). We remark on the following differences. (1) Our approach focuses on the transport decisions involved when using RTIs as a method of packaging, analysing, for example, the method of transport used and transport capacity constraints. (2) We also focus on the integration of RTIs with other subsystems of the supply chain, such as inventory management and production decisions. (3) We only focus on research articles that aim to study RTIs system design and the operational decisions made. This article aims to answer the proposed research questions (RQs):

- (RQ1) What issues are identified in the scientific literature regarding integrating transport planning and using RTIs within the supply chain, and what indicators are used to verify their effectiveness?
- (RQ2) What quantitative models and software tools are used to integrate transport planning and the use of RTIs in a supply chain?
- (RQ3) What are the main research gaps and future trends identified in the scientific literature regarding the integration of transport planning and the use of RTIs within supply chains?

This paper aims to provide a comprehensive literature review on quantitative models related to the transport decisions of returnable transport items (RTI), aiming to enhance understanding of RTI management from both a quantitative and sustainable perspective. Articles were selected, reviewed, and classified based on three key criteria: RTI usage, transport decisions, and quantitative modelling approaches. The main contributions of this review are threefold: (i) it surveys the existing literature; (ii) it categorizes the literature based on various factors such as purpose, context, supply chain structure, decision level, information sharing,

transportation decisions, other subsystems, modeling approaches, solution methods, and software tools; and (iii) it analyzes current trends to highlight potential avenues for future research.

The structure of the paper is as follows. Section 2 presents the review methodology...

2.3 Methodology

Our review methodology follows a structured approach inspired by Díaz-Madroñero et al. (2014) and Becerra et al. (2021). First, the RQs are defined to reach the research objectives.

The second step involves retrieving relevant scientific studies to address the research questions. Table **??** lists the keywords applied during this literature search. Using these terms, we constructed search queries—based on TITLE-ABS-KEY in Scopus and TS in the Web of Science Core Collection—which were executed in both databases as detailed in Table **??**.

Returnable	**Transport Items**	**Transport Decisions**	**Context**
reusable OR returnable OR refillable OR circular	container* OR package* OR packaging OR item* OR box* OR pallet* OR cage* OR crate* OR rack* OR bin*	transport* OR route* OR distribution OR delivery	Supply and Chain

Table 2.1: Search keywords

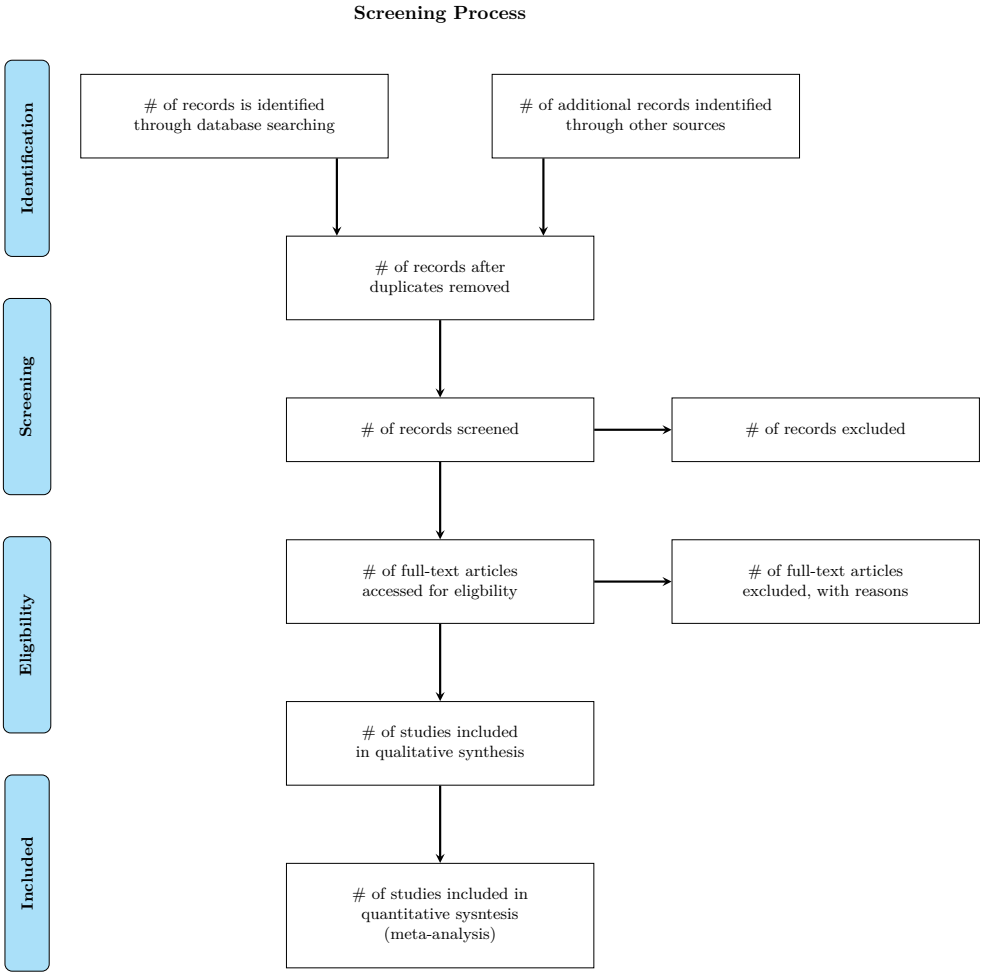
The following image illustrates the selection process of the review.

The third step is to decide which studies from the databases are relevant to answer the research questions. Therefore, the following inclusion criteria are defined:

Inclusion criteria

- The article's main topic is the use of returnable containers in a supply chain.
- The article contains transport decisions.
- The article contains a quantitative modelling approach.

According to the research methodology, the fourth step is to define research classification criteria. We will expand the classification criteria defined by Díaz-Madroñero et al. (2014) by adding three specific criteria: RTI system, transport decision and constraints, integration with other subsystems, such as production or inventory management.


Screening Process
Identification
Screening
Eligibility
Included
# of records is identified through database searching
# of additional records indentified through other sources
# of records after duplicates removed
# of records screened
# of records excluded
# of full-text articles accessed for eligbility
# of full-text articles excluded, with reasons
# of studies included in qualitative synthesis
# of studies included in quantitative sysntesis (meta-analysis)

DataBase	Search String	Number of papers
SCOPUS	TITLE-ABS-KEY (container* OR package* OR packaging OR item* OR box* OR pallet* OR cage* OR crate* OR rack* OR bin*) AND TITLE-ABS-KEY (reusable OR returnable OR refillable OR circular) AND TITLE-ABS-KEY (transport* OR route* OR distribution OR delivery) AND TITLE-ABS-KEY (supply AND chain)	210
WOS	(TS=(container* OR package* OR packaging OR item* OR box* OR pallet* OR cage* OR crate* OR rack* OR bin*) AND TS=(reusable OR returnable OR refillable OR circular) AND TS=(transport* OR route* OR distribution OR delivery) AND TS=(supply AND chain)) AND ((DT==("ARTICLE") AND LA==("ENGLISH")) NOT (SILOID==("PPRN")))	233

Table 2.2: Search strings

Objectives: The objective/s of the quantitative model, such as minimising transportation and inventory costs or maximising benefits and managing RTIs. Consider whether the goal is to minimise or maximise, and note if there are single or multiple objectives.

Application Context: Describe the practical applications each research article considers, including numerical experiments or case studies in real-world supply chains.

Supply-Chain Structure: Categorise the arrangement of various organisations within the supply chain and how they interact with one another.

Decision Level: Time horizon addressed: strategic (long-term), tactical (mid-term), or operational (short-term).

Shared Information: This encompasses the data exchanged between each node in the network as determined by the model, facilitating RTIs management

and transport planning in line with the established objectives.

RTI Management: This involves the study of the RTIs management system, focusing on ownership, responsibilities, and roles of supply chain members, including hubs, cleaning facilities, and warehouses.

Transport Decisions & Constraints: This includes decisions related to transportation, such as cost, method, and capacity constraints.

Inventory Management: This refers to the policies of suppliers, manufacturers, or distributors for replenishing products and RTIs, and details the inventory variables and parameters used in modelling.

Production Decisions: Production-planning rules or decision-support systems (lot sizing, batch scheduling) at manufacturing facilities.

Modelling Approach: This refers to the type of problem representation, specifically the mathematical relationships and the key aspects considered within the supply chain.

Solution Approaches: Algorithms or techniques used (exact methods, heuristics, metaheuristics, simulation) to obtain a solution.

Software Tools: Commercial or open-source software used to implement and solve the models.

Chapter 3

An optimisation approach for supply chain procurement transport operational planning with 3D full-truck loading constraints and returnable transport items

This paper proposes a new mixed integer linear mathematical programming (MILP) model, dubbed 3D-SCPTOP, for procurement transport planning in manufacturing supply chains. The model effectively integrates procurement and transport decisions while accounting for three-dimensional full-truck loading constraints and managing returnable transport items (RTIs). Thus, we optimise the material to procure for an assembly manufacturing site, RTIs closed-loop management, and the involved transport decisions, including the truck loading configuration. Procurement transport decisions are based on the demand of assemblers, the supplier's inventory levels, and the dimensions of the truck / RTIs. The optimisation model aims to minimise transportation and inventory costs. As the 3D loading problem is NP-hard, a matheuristic (Math-3D-SCPTOP) approach is developed to obtain good quality solutions for large instances in an efficient computational time. The 3D-SCPTOP model was validated with a case study in an automo-

tive seating first-tier supplier. Regarding inventory and transport efficiency, the results of the 3D-SCPTOP model are superior to the company procedures.

Keywords— Procurement, inventory, transport planning, full-truckload, 3D container loading

3.1 Introduction

Optimizing the use of resources in manufacturing supply chains is crucial for boosting economic growth, improving social well-being, and minimizing environmental impact. Integrating transport, inventory, and production processes is a fundamental characteristic of supply chains (**Stadtler2015SupplyPlanning**) because joint decisions will improve supply chain performance(**Diaz-Madronero2017ADecisions**).

Procurement refers to the process of acquiring goods and services in a supply chain (**Chopra2015SupplyOperation**). One of the main goals of procurement is to obtain raw materials, parts or finished goods for a factory or distribution centre from a supplier. As mentioned in (**Fleischmann2015TransportDistribution**), the responsibility for transport planning usually lies with the supplier. However, sometimes the manufacturer makes transport decisions of its supplies in, for example, the automotive industry. Furthermore, an essential asset in the automotive supply chain is not considered in traditional material requirement planning (MRP): returnable containers or returnable transport items (RTI), which are used as tertiary packaging. RTIs enable the storage and transportation of RM and FG through the supply chain. Usually, the manufacturing plant supplies empty RTIs to its supplier, while the supplier delivers full RTIs with the FG or RM. Therefore, during the procurement process, transport and inventory decisions should be integrated, including RTI management. Here, we focus on optimising the operational decisions made during procurement transport planning.

According to (**Diaz-Madronero2014AChain**), using algorithm-based methods for procurement transport planning is more effective than relying on manual approaches based on planners' intuition. Manual planning can be short-sighted and lead to less efficient outcomes. This becomes even more significant when optimizing procurement transport tasks that aim to reduce the number of trucks and inventory levels, especially under 3D loading constraints where the precise placement of containers within trucks must also be determined (cite). Furthermore, the capacity constraints used in the scientific literature on procurement transport planning are based mainly on freight volume or the total number of loaded product units (**Hashemi-Amiri2023IntegratedApproach**) and (**Diaz-Madronero2014AChain**). The current research seeks to address this gap by introducing a new mixed-integer linear programming (MILP) model. This model uniquely combines procurement transportation with considerations for 3D loading, inventory management, and RTI management. Hereafter, we refer to this model formulation as 3D supply chain procurement transport operational planning (3D-SCPTOP). Additionally, a matheuristic procedure (MATH-3D-SCPTOP) is researched.

In this paper, we present three main contributions: (i) formulating a novel MILP model, 3D-SCPTOP, that integrates procurement transport, RTI management and inventory decisions with 3D full-truck loading constraints; (ii) developing a matheuristic, Math-3D-SCPTOP, to solve the 3D-SCPTOP model in a reasonable time, which is to be used for real-world instances; (iii) validating and testing the 3D-SCPTOP model with a case study at a first-tier supplier of the automotive sector.

Readers are referred to the typology by (**Wascher2007AnProblems**) for a global view

of container loading problems, where the different dimensions of the loading problem are described. Furthermore, (**Silva2019ExactProblems**) summarise the exact methods used in the literature to solve container loading problems, and (**Zhao2016INTERNATIONALAlgorithms**) explore other solution approaches in the literature. Readers are also referred to the literature reviews of (**Mosca2019IntegratedReview**) and (**Diaz-Madronero2015ADecisions**), which focus more on integrating inventory decisions and transport. Here, we jointly address container loading, procurement transport and inventory decisions. Furthermore, we refer readers () for insights about the packaging methods used in supply chains.

The paper is organised as follows. Section 2 provides a literature review related to the problem under study. Section 3 presents the problem description. Section 4 describes the 3D-SCPTOP MILP formulation and the developed matheuristic solution approach, MATH-3D-SCPTOP. Section 5 validates and experiments with the proposal in a real case study. Finally, Section 6 concludes and puts forward further research lines.

3.2 Literature Review

The goal of this research is to integrate procurement transport operational planning decisions with 3D loading restrictions and RTI packaging. Based on the authors' research, no previous work has been identified that meets this criterion. Therefore, three independent, but sometimes overlapping, knowledge bodies have been reviewed: procurement transport and inventory planning, container loading problems and RTI literature.

3.2.1 Procurement transport and inventory planning

Models that integrate transport and inventory costs during the procurement process are widely studied in the literature. As noted in (**Langley2020SupplyPerspective**), transportation facilitates the movement of raw materials and finished products between suppliers, warehouses, manufacturers, and retailers. Therefore, transport processes play a crucial role in supply chains. (**Fleischmann2015TransportDistribution**) distinguish between two main types of transportation within a supply chain: "procurement transport," which involves moving materials from suppliers to manufacturing sites, and the "distribution of goods," which concerns delivering finished products from factories to customers. This article focuses specifically on addressing the challenges related to procurement transportation.

There are two main dimensions of procurement transport problems according to the supply chain configuration and the applied transport method. Regarding supply chain configurations (**Mosca2019IntegratedReview**): (i) single source to a single destination (SS); (ii) multiple sources to a single destination (MS); (iii) single source to multiple destinations (SM); (iv) multiple sources to multiple destinations (MM). Source components determine the number of supplier locations that are taken into account, while destination components denote the number of supplied manufacturers. Additionally, different transport methods are identified in (**Diaz-Madronero2014AChain**):milk run, less-than-truckload (LTL), full-truckload (FTL) . We also analyze the formulation of the transportation capacity constraints in the scientific literature.

(**Berman2006InboundCost**) address the problem of optimising distribution strategies to deliver products from suppliers to plants, whose aim is to minimise transport, pipeline and plant inventory costs by using cross docks. A mixed integer non-linear programming (MINLP) formulation, a Lagrangian relaxation heuristic, a greedy heuristic, and a branch-and-bound algorithm are proposed to model and solve the problem.

(**Sancak2011Multi-itemPolicy**) present a MILP model to optimise procurement trans-

port decisions in a SS supply chain configuration using LTL and FTL transport methods. The objective is to minimise transport and inventory holding costs, while fulfilling production plan requirements on a multiperiod planning horizon. The study explores the possibility of delaying LTL shipments to the subsequent time period, and utilising safety stocks whenever necessary, in a stochastic environment by employing data from a bus manufacturer. The findings demonstrate that delaying shipments leads to reduced holding and transport costs with a minimal stockout risk.

(**Schoneberg2010AnNetworks**) examine the concept of delivery profiles, which define the delivery frequency at a fixed rate, offering structured planning, minimizing management effort, and improving coordination in logistics. In this context, FTL transport is used when the goods' volume exceeds a vehicle's capacity, making direct shipment possible. LTL is applied when the volume is insufficient to fill a truck, and the goods are first sent to a consolidation center before reaching the final destination. In (**Schoneberg2013ANetworks**), the proposed model accounts for demand uncertainty.

(**Diaz-Madronero2014APlanning**) present a fuzzy multi-objective integer linear programming (FMOILP) model that focuses on operational procurement transport planning in an automotive supply chain, which involves a single source and a single destination while utilising FTL. The developed model aims to minimise the number of trucks employed for procurement transport and the inventory level by considering the uncertainties related to transport capacity levels.

(**Hein2016QuantitativeProblem**) investigate the benefits of integrated procurement transport and production planning for supply and production processes by comparing a sequential approach, where production is firstly determined, followed by procurement transport decisions, plus an integrated approach in which decisions are jointly made. Numerical experiments reveal that the integrated planning approach leads to substantial cost savings.

(**Diaz-Madronero2017ADecisions**) present an novel MILP model that integrates procurement, production and transport planning decisions within manufacturing systems. This model extends beyond traditional material requirement planning (MRP) systems by incorporating procurement transport decisions from multiple sources and a single destination into a unified optimization framework. It considers the capacities of both material and production resources, along with procurement transport planning decisions, by employing various shipping modes such as FTL, LTL, and milk-run.

(**Chitsaz2019ARouting**) address integrating production, inventory and procurement transport into a manufacturing context by defining this problem as an assembly routing problem (ARP). It introduces a novel MILP model for the ARP and proposes a three-phase decomposition matheuristic for its solution.

(**Baller2022OptimizingApproach**) examine the optimization of procurement transport in an automotive factory, focusing on first-tier suppliers at an aggregate level across multiple sources (570) to a single destination, using FTL, LTL, and courier express service (CES) as transportation methods. The proposed MILP approach includes a cyclic inventory constraint, truck utilization, and storage limitations.

(**Hashemi-Amiri2023IntegratedApproach**) propose an integrated approach to optimize a perishable food supply chain. The model combines supplier selection, procurement and distribution transport, production scheduling, and vehicle routing to improve supply chain efficiency and resilience. It handles demand uncertainties through a distributionally robust chance-constrained model. The model is reformulated as MILP by utilizing duality and linearization techniques.

3.2.2 Procurement Transport with RTIs

Some models further integrate the procurement and delivery of RTIs into the procurement transport literature. In this sense, RTIs are procured to the supplier, who loads the items and sends them to customer. We extend the SC definitions previously defined to contemplate the RTIs management strategies.

model a food closed-loop supply chain network design problem incorporating returnable transport items (RTIs), product perishability, facility location and demand uncertainty. A two-stage non-linear chance-constrained programming model minimizes fixed facility, production, transport, inventory holding and RTI costs while ensuring a target service level. It's linearized via sample average approximation and distribution-free methods to capture forward RTI-based shipments and reverse flows for reuse. To solve large instances, an improved Lagrangian relaxation algorithm decomposes facility-location and flow decisions, outperforming CPLEX on solution quality and speed.

(**SOYSAL2023101504**) develop a tactical-level, multi-period closed-loop Vendor-Managed Inventory (VMI) system for RTIs in an agro-food SC. The SM SC integrates forward flows of full RTIs from a single vendor to multiple customers and reverse flows of empties back to the vendor for cleaning . All routing and inventory decisions are made centrally by the vendor under full information sharing. Sustainable plans in terms of amounts of energy spent and empty crates collected are devised and evaluated.

(**GUO2024104339**) model two-tier Container Management Centre (CMC) network . Empty RTIs are dispatched from central CMC hubs to upstream suppliers, loaded with products, carried by truck to downstream customers, then returned empty to regional CMCs for cleaning and storage before being shipped back to central hubs. A neural network is used for demand forecasting. A MILP is proposed that maximises total profit by optimally determining transport costs and the penalties applied to late RTIs returns.

(**Zhang17012024**) formulate a perishable-food inventory-routing MILP with multiple RTI types in a SM supply chain, integrating procurement of products and pickup of both filled and empty RTIs. Decisions are made on a operational level, optimally determining per period the quantities to pack, deliver, pick up, purchase and lose, as well as the vehicle routes in order to maximise the total profit of the SC that is equal to the total quality-dependent revenue minus the total costs involved, including inventory holding costs, product cost, transportation costs, and RTI purchasing and loss costs. The model is solved exactly via CPLEX solver and approximately via a tailored Kernel Search matheuristic.

3.2.3 Container loading

Container loading problems encompass a subset of geometric assignment problems known as cutting and packing ($C\&P$) problems. They revolve around the allocation of n-dimensional orthotope items in larger n-dimensional orthotope containers. Here the aim is to optimise an objective function while taking into account two key geometric constraints: complete containment of items inside containers and lack of item overlap. A loading pattern serves as a formal representation of a solution to such an assignment problem. Dyckhoff introduced in (**Dyckhoff1990AProblems**) a typology that unifies $C\&P$ problems using four different criteria: dimensionality of space, the objective of assignment, containers types and box type. Consequently, (**Wascher2007AnProblems**) develop a new typology that has become the standard in the $C\&P$ literature. We briefly describe the typology used in (**Wascher2007AnProblems**) based on five different criteria: (i) dimensionality of space; (ii) kind of assignment (input minimisation of containers for all small items and loading

patterns; output maximisation to pack a subset of small items with limited container availability;) (iii) assortment of small item by distinguishing if they are either a weakly (or strongly) heterogeneous assortment or identical small items ; (iv) assortment of large objects (if it is either several large objects or one large object); (v) shape of small items (circular or of another shape). Table 1 links the standard $C\&P$ nomenclature and the criteria of the typology according to (**Wascher2007AnProblems**).

Table 3.1: $C\&P$ nomenclature and criteria

<table>
<tr><td colspan="2">Assortment of small items</td><td>Identical</td><td colspan="2">Weakly hetereogenous</td><td colspan="2">Strongly heterogenous</td></tr>
<tr><td colspan="2">Objective</td><td>Output maximization</td><td>Input minimization</td><td>Output maximization</td><td>Input minimization</td><td>Output maximization</td></tr>
<tr><td colspan="2">Assortment of large items</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="4">Fixed Dimensions</td><td>Identical</td><td rowspan="3">X</td><td>Single Stock Size Cutting Stock Problem (SSSCSP)</td><td>Multiple Identical Large Object Placement Problem (MILOPP)</td><td>Single Bin Size Bin Packing Problem (SBSBPP)</td><td>Multiple Identical Knapsack Problem (MIKP)</td></tr>
<tr><td>Strongly heterogeneous</td><td>Residual Cutting StockProblem (RCSP)</td><td rowspan="2">Multiple Heterogenous Large Object Placement Problem (MHLOPP)</td><td>Residual Bin Packing Problem (RBPP)</td><td rowspan="2">Multiple Heterogeneous Knapsack Problem (MHKP)</td></tr>
<tr><td>Weakly hetereogenous</td><td>Multiple Stock Size Cutting Stock Problem (MSSCSP)</td><td>Multiple Bin Size Bin Packing Problem (MBSBPP)</td></tr>
<tr><td>One large object</td><td>Identical Item Packing Problem (IIPP)</td><td>X</td><td>Single Large Object Placement Problem (SLOPP)</td><td>X</td><td>Single Knapsack Problem (SKP)</td></tr>
<tr><td>Variable dimension</td><td>One large object</td><td>X</td><td>Open Dimension Problem (ODP)</td><td>X</td><td>Open Dimension Problem (ODP)</td><td>X</td></tr>
</table>

The first mathematical programming model in a scientific journal to describe the container-loading problem was developed in (**Chen1995EUROPEANProblem**), where the authors introduced an MILP model to address the 3D input minimisation problem by considering rotations and containers of varying dimensions. However, it does not specify the use of weakly or strongly heterogeneous boxes or containers. Additionally, real-world constraints are modelled like, for example, weight distribution.

(**Junqueira2012Three-dimensionalConstraints**) propose an alternative MILP formulation to (**Chen1995EUROPEANProblem**) using binary encoding for box position, which utilises normal patterns to reduce possible box coordinates by considering multiples of boxes dimensions and ensuring optimal solutions. Additionally, loading and cargo constraints are developed to ensure the static and dynamic stability of the cargo. Boxes must be supported by other box top faces or the container floor. The same authors integrate further loading restrictions in (**Junqueira2012MIP-basedConstraints**), where they integrate multidrop constraints, which address the challenge of packing a truck with boxes for different customers at various destinations. The goal is to plan the container's loading to minimise the time spent unloading and reloading boxes at each destination.

(**Zhu2012AProblem**) devise a tree search-based method to solve SLOPP and SKP with 3D constraints. Their approach involves evaluating free spaces by employing a look-ahead tree search and selecting optimal placements based on the best free space-block pairs.

(**Goncalves2013AProblems**) develop a biased random-key genetic algorithm (BRKGA) designed to solve SBSBPP. The algorithm addresses geometric constraints and introduces

a maximal-space representation to handle available free spaces in bins. A novel placement procedure is hybridised with the BRKGA, which optimises the order of box packing and the parameters utilised by the placement procedure to offer an effective solution approach to an NP hard problem.

(**Bortfeldt2013ConstraintsReview**) identify constraints for $C\&P$, including item-related constraints (orientation constraints, load bearing, loading priorities), container-related constraints (weight distribution and weight limit), and cargo-related constraints (complete-shipment constraints) and load-related constraints (complexity constraints or stability constraints). The study also highlights the achievements and deficits in container loading research that pertain to these constraints.

(**2016Paquay**) propose an MILP model for the MBSBPP. One noteworthy constraint is the assumption that containers are truncated parallelepiped instead of rectangular boxes. They also consider constraints, such as box rotation, container load bearing, weight distribution, stability and fragility of items.

(**Mahvash2018**) introduce a novel resolution procedure for the SBSBPP through the column generation technique, where the columns that are generated are the loading configurations of a single bin. For the pricing develop problem they develop a heuristic to find new columns with a negative reduced cost. This approach not only produces better results than the previous articles on the literature, but also gives new lower bounds.

(**Silva2019ExactProblems**) study SSLOP models with 3D constraints. These authors compare the different approaches that exist in the literature for model packing problems based on two fundamental approaches: discretised formulation (**Junqueira2013AnConstraints**) and continuous formulation (**Chen1995EUROPEANProblem**) with subsequent modifications. They also indicate that discretised formulation(**Junqueira2012MIP-basedConstraints**) performs the best for instances with similar sized boxes. However, it encounters memory limitations for large instances. For other sets of instances, the adapted formulations of (**Chen1995EUROPEANProblem**) and (**2016Paquay**) outperform other adaptations. Further research topics suggest including the design of new formulations by considering the limited approaches in the literature.

(**Kurpel2020TheProblems**) introduce techniques and exact approaches to solve input minimisation and output maximisation problems for container-loading problems. Their methods consider practical constraints, such as orientation, separation of boxes and vertical stability. A set partitioning method is also employed in order to obtain upper bounds in a container-loading input minimisation problems. The authors adapt and evaluate four discretisation methods found in the existing literature and incorporate mathematical formulations to address the aforementioned constraints.

(**Alonso2020AConstraints**) develop an algorithm by the greedy randomised adaptive search procedure (GRASP) to tackle a practical container-loading problem with time-sensitive constraints. Their approach addresses dynamic stability constraints while stacking products on pallets and loading them onto trucks during different time periods. Their approach provides high-quality solutions and surpasses an integer programming formulation in efficiency and effectiveness terms.

(**Ali2022On-lineApproaches**) conduct a comprehensive literature review by identifying constraints and solution methods for offline problems with full knowledge of the items to be packed and those tackling the online problems, where items are packed immediately when they arrive, and no prior knowledge of an item is known. The study also identifies and classifies solution approaches for offline packing into six categories: exact methods, constructive heuristics; local search and metaheuristic methods; approximation algorithms;

machine learning and hyper-heuristic methods; tree search methods. (**Harrath2022**) proposed a thee stage heuristic based on layer generation to find good quality solutions the SBSPP problem for large instance up to 993 bins and 200000 items taking into account load balancing constraints.

(**Tresca2022**) develop a matheuristic to solve the SBSBPP that includes items with different IDs and different categories, to be delivered to different clients. The algorithm creates monoitem, monocategory, and mixed layers which then are used to compute the final loading configurations of the bins.

(**Li2022**) develope a hybrid adaptaive large neighborhood search algoritm to solve the multiple container loading problem, which is tries pack a given set of items with the minimum number of containers while maximazing the space utilization of them. Therfore, it is both a MBSBPP and a MSSCSP. Their approach takes into account some practical applications such as supsension constraints, rotation and weight limit.

(**Chen2023**) present a new approach to the 3D-BPP, introducing the 3D-BSDPP which focuses on optimizing both the design of bin sizes and the packing process. A novel Hybrid Biogeography-Based Optimization (HBBO) algorithm is proposed, integrating differential evolution and a local search strategy for enhanced performance.

One noteworthy trend in $C\&P$ is integrating loading problems with routing. This is known as capacitated vehicle-routing problem with 3D or 2D loading constraints (3L-CVRP or 2L-CVRP). (**Junqueira2012Three-dimensionalConstraints**) present a MILP model with binary encodings for 3L-CVRP by including loading, cargo and multidrop constraints. The combination of these two NP-hard problems requires using heuristics and metaheuristics for real-world instances, as seen in (**Dominguez2016UsingFleet**), (**Alonso2022TheRevision**) and (**Wei2015AConstraints**). Another noteworthy trend is to use reinforcement learning techniques to solve $C\&P$, as in (**Que2023SolvingLearning**) and (**Zhao2022LearningPacking**).

3.2.4 Research Gaps & Methodology

Here, we discuss the main differences between the 3D-SCPTOP model proposal and those reviewed by specifically focusing on procurement transport. The key distinction of the 3D-SCPTOP model compared to the existing literature is the incorporation of 3D constraints and vertical stability constraints to model transport capacity with a view to providing more realistic solutions and reducing the likelihood of infeasible outcomes. Table **??** presents a comparison of inventory modelling, transport modelling, and supply chain configuration between the 3D-SCPTOP model proposal and those currently in the literature. Hence, the 3D-SCPTOP approach is based on a SS supply chain configuration, and takes into account the FTL transportation mode.

On the other hand, the 3D-SCTOP model diverges from those proposed in the packaging literature in several aspects. Firstly, the loading problem takes into account different time periods. Therefore, the specific container (small items or boxes in the $C\&P$ literature) that we utilise are yet to be defined, as the choice of container depends on the inventory levels of each product. Also, the objective function depends on inventory and transport costs, which are inherently connected to the loading problem at hand. Thefore, we simultaneously solve an input minimisation and output maximisation problem. Both of these are a result of transport minimisation and inventory costs. Input minimisation comes because we want to find the minimum number of trucks needed. Meanwhile, output maximization comes from the FTL and the transport costs.

Based on the literature review and the identified research gaps, our proposal is structured according to the steps depicted in Figure **??**.

Article	SC Configuration	Decision level	Transportation Mode	Transportation Capacity Constraint	Transportation Cost	Inventory Capacity Restriction	RTIs	Inventory Costs	Safety Stocks	Modelling Framework
(**GUO2024104339**)	MM	TO	FTL	UH	X	X	S|NF	X		MINLP
(**Berman2006InboundCost**)	MM	T	FTL|LTL	V	X			X		MINLP
(**Sancak2011Multi-itemPolicy**)		O	LTL|FTL	V	X			X	X	MILP
(**Schoneberg2010AnNetworks**)	MS	T	FTL|LTL	V	X	X		X		MILP
(**Schoneberg2013ANetworks**)	MS	O	FTL|LTL	V	X	X		X		MILP
(**Diaz-Madronero2014APlanning**)	SS	O	FTL	UH	X			X	X	FMOILP
(**Hein2016QuantitativeProblem**)	MS	O	MR	V	X			X		MILP
(**Diaz-Madronero2017ADecisions**)	MS	T	FTL|LTL|MR	V	X	X		X	X	MILP
(**Meyer2018TransportIndustry**)	MS	T|O	MR|FTL	V	X	X		X		MILP
(**Chitsaz2019ARouting**)	MS	O|T	MR	V	X	X		X		MILP
(**Baller2022OptimizingApproach**)	MS	O	MR|FTL|LTL|CES	V|W	X	X			X	MILP
(**Hashemi-Amiri2023IntegratedApproach**)	MS	T	MR	V	X					CCP
3D-SCPTOP	SS	O	FTL	3D-Constraints	X	X		X	X	MILP
(**Zhang17012024**)	MS	TO	MR	UH	X	X	M|NF	X		ILP

Abbreviations. T (tactical), **O** Operational; **FTL** (full-truck load), **LTL** (less-than-truck load), **MR** (milk-run), **CES** (courier express service); **V** (volumetric), **W** (weight), **UH** Unit Holding; **CCP** chance-constrained programming

Table 3.2: Comparison of procurement transport models

Article	Problem type	Resolution procedure	Orientation	Weight	Load Balance	Stability	Time-Depenent
(**Zhu2012AProblem**)	SLOP|SKP	TSM	X	X			
(**Goncalves2013AProblems**)	SBSBPP	BRKGA	X	X		X	
(**2016Paquay**)	SBSBPP	EM	X	X	X	X	
(**Mahvash2018**)	SBSBPP	CG	X	X			
(**Alonso2020AConstraints**)	SSSCSP	GRASP	X	X	X	X	X
(**Tresca2022**)	SBSBPP	MH	X	X	X	X	
(**Harrath2022**)	SBSBPP	H	X	X	X		
(**Li2022**)	SBSBPP|MSSCSP	HALNs	X	X		X	
(**Chen2023**)	BSDPP	HBBO	X	X			
3D-SCPTOP	SSSCSP & MILOPP	MH	X	X	X	X	X

Abbreviations. TSM (Tree search method), **BRKGA** Biased Randomized Jey Genetic Algorithm; **EM** (Exact Metods), **MH** (MatHeuristic), **CG** (Column Generation), **H** (Heuristic); **HCLP** (Hybrid Adaptive Large Neighborhood Search), **HBBO** (Hybrid Biogeography-Based Optimization)

Table 3.3: Comparison of C&P problems

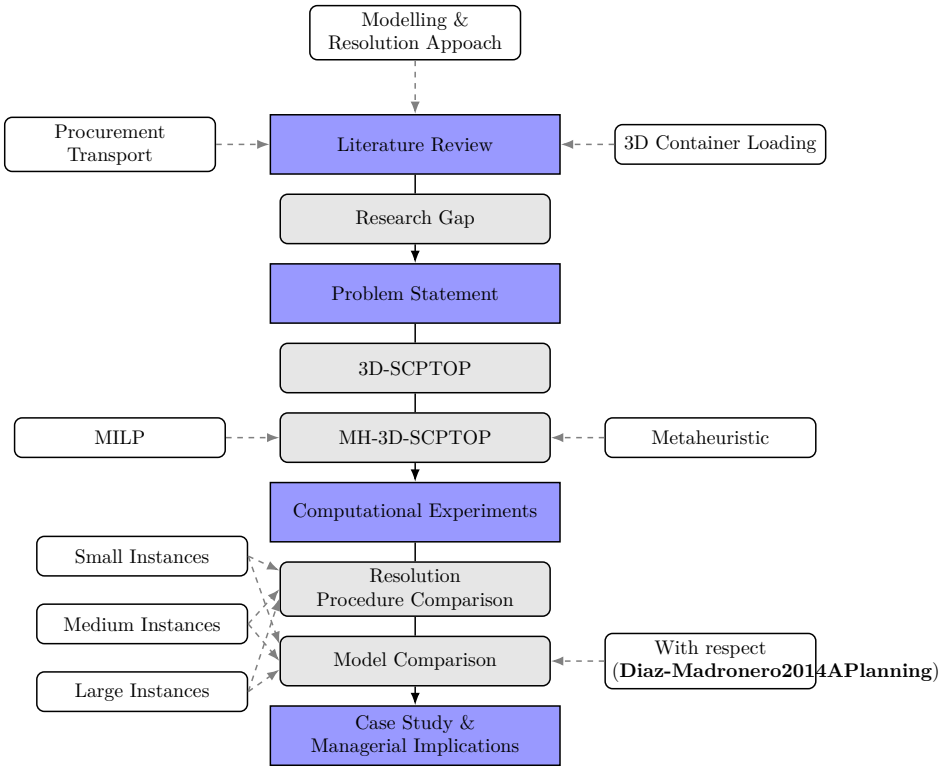


Figure 3.1: Research Methodology

3.3 Problem Description

The main objective of the presented problem is to minimise the transportation and inventory holding costs of a manufacturing site by integrating procurement transport, RTI management and inventory decisions with 3D full-truck loading constraints. The supply chain under study has a single source and a single destination. FTL is employed as the mode of transport. Products are shipped in different RTI types. RTIs are transported inside trucks, and the loading configuration is determined. Therefore, a distinction is made to the classic nomenclature of $C\&P$ problems. Hereafter, small objects are RTIs and large objects are trucks.

The assembler's demand drives the procurement process because it determines the optimal quantity of the products to procure from the supplier to the manufacturing site. Raw materials and assembled parts are produced at the supplier following the production plan. Then, they are packed in their corresponding RTI. The full RTIs with RM and AP are transported using an FTL mode of transport with 3D loading constraints. Each RTI can only carry the same product type at a given time. Furthermore, each RM or AP can only be shipped in one RTI type. Once the full RTIs are at the plant, the RM and AP are used for production, and the RTIs are emptied. Once emptied, they are folded and shipped back to the supplier using FTL transport with 3D loading constraints. Figure **??** is an example of the assignment of RM and AP to RTI types, where three container types and six product types are represented in different colours. The arrow indicates the amount of product that can be shipped in the containers.

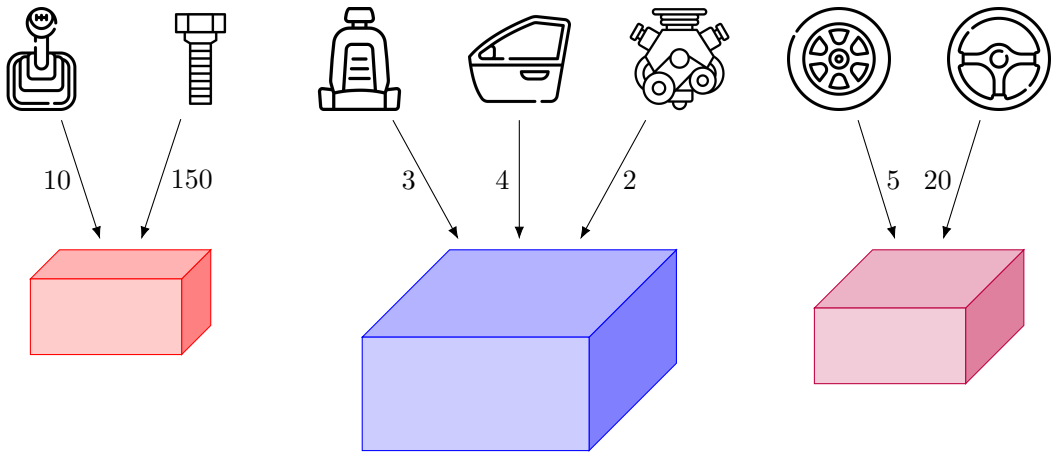


Figure 3.2: Product to type of reusable container

In our problem, two distinct transport decisions are made: procuring RM and AP for the manufacturing plant with full RTIs. And the supply of empty RTIs to the supplier.

Several constraints and assumptions are considered in this model formulation. Firstly, the physical transport constraints ensure that RTIs do not overlap and are contained in the truck. A truck's maximum weight limit must not be exceeded. Maintaining the static stability of containers during transport is crucial to prevent tipping or shifting because it

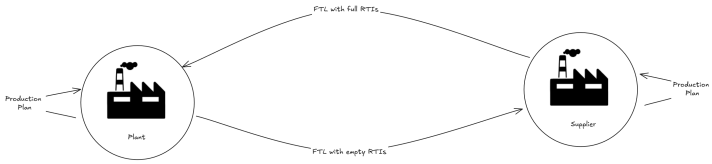


Figure 3.3: Enter Caption

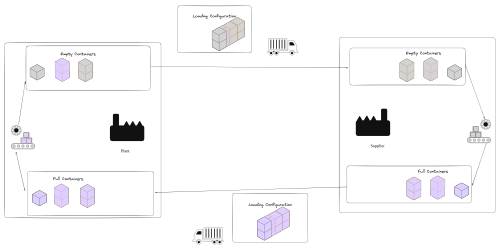


Figure 3.4: Enter Caption

could damage RM and AP. Rotations of containers about z-axes are allowed. As FTL is the mode of transport employed, trucks should be loaded to achieve a minimum utilisation percentage. Furthermore, we assume that the supplier's production plans and the plan are synchronised to guarantee the production of the manufacturing site. Minimum inventory coverage constraints are considered, subject to the product's future demand. Thus, the problem can be formulated in the following way:

Initial data:

- Product data: initial inventory, demand for each time period of the plant in terms of full RTIs, the type of reusable container in which it must travel, production demand for each time period at the supplier in terms of RTIs, and the weight of the RTI when loaded with the given product
- Reusable container data: RTI dimensions when full and folded.
- Transport data: number of available trucks, minimum utilisation percentage, dimensions and maximum weight

To determine:

- The number of trucks used daily with their corresponding loading configuration.
- The inventory level in terms of full RTIs for each product in the factory and supplier, given that everything goes according to plan.
- The inventory level of folded RTIs in the factory and supplier.

Subject to:

- Loading restrictions (i,e: containers cannot overlap, rotations about z axes are allowed, the maximum weight of a truck can not be exceeded, static stability)
- Minimum coverage stock

In order to:

- Minimise the total transportation cost
- Minimise the total inventory holding cost

3.4 Mathematical model

The 3D-SCPTOP mathematical programming model integrates container loading constraints and procurement transport decisions. It differs from traditional 3D container packing problems in two significant aspects. Firstly, the quantity of containers required for loading depends on product demand and the inventory levels of the manufacturing site. Therefore, the specific products to be transported are identified to minimise transport and inventory costs. As inventory levels depend on past loading decisions, a time dimension has to be taken into account. Secondly, it combines output maximisation problems and input minimisation. On the one hand, the number of trucks to be used is minimised, i.e. input minimisation. On the other hand, trucks must be fully loaded, i.e. output maximisation. It also extends traditional procurement transport planning approaches by taking into account 3D loading constraints and RTI management.

Table **??** presents the nomenclature of the 3D-SCPTOP model.

Table 3.4: Nomenclature of 3D-SCPTOP

Indices	
R	Set of RTI types ($r = 1, \ldots, R$)
I	Set of RTIs ($i = 1, \ldots, I$)
K	Set of trucks ($k = 1, \ldots, K$)
N	Set of products ($n = 1, \ldots, N$)
L	Set of loading configurations ($l = 1, 2, \ldots, L$)
T	Set of planning periods *(t =1, ... T)*
G	Set corresponding to the supplier and plant nodes $G = \{s, p\}$
Set of indices	
$M(r)$	Set of RM or AP that are carried in RTI $r \in R$, $n = 1, \ldots, M(r)$
Parameters	
cK	Cost of using a truck
lt	Lead transport time between the plant and the supplier and viceversa.
i_{gn}	Initial inventory level of $n \in N$ in $g \in G$ in terms of full RTIs
ie_{gr}	Initial inventory level of empty RTIs $r \in R$ in $g \in G$ RTIs
ic_{gn}	Inventory cost of $n \in N$ at $g \in G$ in terms of full RTIs
d_{gnt}	Production plan of $n \in N$ at $g \in G$ during time period $t \in T$
q_{rt}	Initial quantity of empty RTIs $r \in R$ arriving at supplier $s \in G$ in time period $t \in (1, \ldots, lt)$
q_{nt}	Initial quantity of full RTIs of $n \in R$ arriving at p in time period $t \in (1, \ldots, lt)$
lc_{lrg}	Number of containers of type c used in loading configuration l arriving at $g \in G$
ss_{gnt}	Security stock for product $n \in N$ in time period $t \in T$ in $g \in G$ in terms of full RTIs
ss_{gtr}	Security stock of empty RTIs $r \in R$ in time period $t \in T$ in $g \in G$
w_n	Weight of a container carrying product n
pK	Maximum weight capacity of a truck

pK	Maximum weight capacity of truck a truck
Γ_c	Volume of a container type c
Γ	Volume of a truck
α	Minimum volume utilisation of a truck
M	A large enough number to make the logic of the model work;
p_{gr}, q_{gr}, r_{gr}	length, width, and height of a container of type c;
pK, qK, rK	Length, width and height of a truck;
Decision Variables	
I_{gnt}	Inventory of product $n \in N$ at the end of $t \in T$ time period at $g \in G$ in terms of full RTIs
Ie_{gtr}	Inventory of empty RTIs $r \in R$ at the end of $t \in T$ time period at $g \in G$
Y_{gkt}	Binary variable, if one then truck $k \in K$ is used in time period $t \in T$ arriving at $g \in G$ and zero otherwise
Y_{gklt}	Binary variable, if one then truck $k \in K$ is used in time period $t \in T$ departing from $g \in G$ using loading configuration l, 0 otherwise
Q_{nt}	Integer variable indicating the number of RM or AP $n \in N$ in transport during time period t to the manufacturing plant p
Q_{rt}	Integer variable the number of empty RTIs $r \in R$ in transport during time period t to the supplier s
V_{gklrt}	Number of RTIs $r \in R$ in truck $k \in K$ using loading configuration $l \in L$ at time period $t \in T$ departing from $g \in G$
V_{kglnt}	Number of full RTIs with RM or AP $n \in N$ in truck $k \in K$ using loading configuration $l \in L$ at time period $t \in T$ departing from $g \in G$
I_{nt}	Inventory of product n at the end of time period t
X_{gil}	Binary variable indicating whether container i is used in loading configuration l
X_{gilr}	Binary variable indicating whether container i is going to be of type c for loading configuration l
V_{gklrt}	Number of containers of type c in truck k using loading configuration l at time period t
V_{kglnt}	Integer variable indicating the quantity of containers carrying product n supplied on truck k using loading configuration l at time period t.
PX_{gil}, PY_{gil}, PZ_{gil}	Coordinates of the lower left front corner of container i on the truck k during planning period t;
LX_{gilr}, LY_{gilr} , LZ_{gilr}	Defines whether the length of container i of type c is on the same direction as the x-, y- or z-axis in loading configuration l. For example, LX_{gilr} equals one if the length of container i of type c is is on the same direction as the x-axis, otherwise LX_{gilr} equals 0;
WX_{gilr} , WY_{gilr} , WZ_{gilr}	Defines whether the width of container i of type c is on the same direction as the x-, y- or z-axis on truck k axis during planning period t. For example, WX_{gilr} equals 1 if the width of container i of type c is is on the same direction as the x-axis, otherwise WX_{gilr} equals 0;
HX_{gilr}, HY_{gilr} , HZ_{gilr}	Defines whether the height of container i of type c is on the same direction as the x-, y- or z-axis on the truck k axis during planning period t. For example, HX_{gilr} equals 1 if the width of container i of type c is is on the same direction as the x-axis, otherwise HX_{gilr} equals 0

A_{gijl}, B_{gijl}, C_{gijl}, D_{gijl}, E_{gijl}, F_{gijl}	Binary variables indicating the relative position between two boxes. Variable A_{gijl} equals 1 if container i is to the left of container j on truck k during planning period t. Similarly, variables B_{gijl},C_{gijl}, D_{gijl} , E_{gijl} , F_{gijl} indicate whether container i is to the right, behind, in front, on top or below container j, respectively, on truck k during planning period t. These variables are only necessary when $i \neq k$.

We wish to minimise the total cost, which is given by transportation and inventory costs.

$$Min\ z = \sum_{g\in G}\sum_{k\in K}\sum_{t\in T} Y_{gkt}\cdot cK + \sum_{g\in G}\sum_{n\in N}\sum_{t\in T} I_{gnt}\cdot ic_{gn} \tag{3.1}$$

(**??**) - (**??**) define the inventory level of RM or AP according to the production demand and supply of RM or AP from the supplier s at the plant p

$$I_{pnt} = I_{pnt'} - d_{pnt} + Q_{nt'} \qquad \forall n \in N, t \in T : t \geq lt+1, t' := t - lt \tag{3.2}$$

$$I_{pnt} = I_{pnt'} - d_{pnt} + q_{nt} \qquad \forall n \in N, t \in (2,\ldots,lt) :, t' := t - lt \tag{3.3}$$

$$I_{pnt} = i_{pn} - d_{pnt} + q_{nt} \qquad \forall n \in N : t := 1 \tag{3.4}$$

$$Ie_{ptr} = Ie_{pt'r} + \sum_{n\in M(r)} d_{pnt} - Q_{rt} \qquad \forall r \in R, t \in T : t \geq 2 \tag{3.5}$$

$$Ie_{ptr} = ie_{pr} + \sum_{n\in M(r)} d_{pnt} - Q_{rt} \qquad \forall r \in R, t := 1 \tag{3.6}$$

(**??**) - (**??**) the empty RTIs according to the production demand and supply of RTIs from the plant p at the supplier s

$$I_{snt} = I_{snt'} + d_{snt} - Q_{nt} \qquad \forall n \in N, t \in T, t' \in T, : t \geq 2, t' := t - lt \tag{3.7}$$

$$I_{snt} = i_{sn} + d_{snt} - Q_{nt} \qquad \forall r \in R : t := 1 \tag{3.8}$$

$$Ie_{str} = Ie_{st'r} + Q_{rt'} - \sum_{n\in M(r)} d_{snt} \qquad \forall r \in R, t \in T : t \geq lt+1, t' := t - lt \tag{3.9}$$

$$Ie_{str} = Ie_{st'r} + q_{rt} - \sum_{n\in M(r)} d_{snt} \qquad \forall r \in R, t \in (2,\ldots,lt) :, t' := t - lt \tag{3.10}$$

$$Ie_{str} = ie_{pr} + q_{rt} - \sum_{n\in M(r)} d_{snt} \qquad \forall r \in R : t = 1 \tag{3.11}$$

Each product must satisfy a minimum security stock ss_{gnt}.

$$I_{gnt} \geq ss_{gnt} \quad \forall g \in G, n \in N, t \in T \tag{3.12}$$

(**??**) imposes that a truck k can only use one loading configuration l with direction $g \in G$ in time period t.

$$\sum_{l\in L} Y_{gklt} = Y_{gkt} \quad \forall g \in G, k \in K, t \in T \tag{3.13}$$

$$X_{gil} = \sum_{r \in R} X_{gilr} \quad \forall g \in G, i \in I, l \in L \tag{3.14}$$

$$V_{glr} = \sum_{i \in I} X_{gilr} \quad \forall g \in G, r \in R, l \in L \tag{3.15}$$

(**??**) ensures that a truck is used if it is carrying containers. Meanwhile, (**??**) - (**??**) impose that the truck k at time period t using loading configuration l uses the same loading configuration as l.

$$V_{gklrt} \leq M(1 - Y_{gklt}) \qquad \forall g \in G, r \in R, k \in K, t \in T \tag{3.16}$$

$$V_{gklrt} + M(1 - Y_{gklt}) \geq V_{glr} \qquad \forall g \in G, r \in R, k \in K, t \in T \tag{3.17}$$

$$V_{gklrt} \leq V_{glr} + M(1 - Y_{gklt}) \qquad \forall g \in G, r \in R, k \in K, t \in T \tag{3.18}$$

Since a product can only be carried by one container type (**??**) sets that the number of containers of type c carried at truck k using loading configuration l at time period c is equal to the sum of containers carried of products n which containers of type c can carry. Moreover, (**??**) sets the total amount of product n being transported from s to p in time period t, and (**??**) sets the total amount of RTIs r being transported from p to s in time period t,

$$V_{pklrt} = \sum_{n \in M(r)} V_{pklnt} \qquad \forall k \in K, l \in L, t \in T, r \in R \tag{3.19}$$

$$Q_{nt} = \sum_{l \in L} \sum_{k \in K} V_{kslnt} \qquad \forall n \in N, t \in T \tag{3.20}$$

$$Q_{rt} = \sum_{l \in L} \sum_{k \in K} V_{pklrt} \qquad \forall r \in R, t \in T \tag{3.21}$$

(**??**) guarantees that the maximum weight allowed for truck k is not surpassed.

$$\sum_{l \in L} \sum_{n \in N} w_n V_{kglnt} \leq pK \qquad \forall k \in K, t \in T, g \in G \tag{3.22}$$

We ensure that two containers are not overlapped by their relative positions. (**??**) and (**??**) to identify whether container i is to the left or the right of container j.

$$PX_{gil} + \sum_{r \in R} (p_{gr} LX_{gilr} + q_{gr} WX_{gilr} + r_{gr} HX_{gilr}) \leq PX_{gjl} + (1 - A_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{3.23}$$

$$PX_{gjl} + \sum_{r \in R} (p_{gr} LX_{gjlr} + q_{gr} WX_{gjlr} + r_{gr} HX_{gjlr}) \leq PX_{gil} + (1 - B_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{3.24}$$

Similarly, (**??**)-(**??**) determine whether one container is ahead, behind, on top of or below one another.

$$PY_{gil} + \sum_{r \in R} (q_{gr} WY_{gilr} + p_{gr} LY_{gilr} + r_{gr} HY_{gilr}) \leq PY_{gjl} + (1 - C_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{3.25}$$

$$PY_{gjl} + \sum_{r \in R} (q_{gr} WY_{gjlr} + p_{gr} LY_{gjlr} + r_{gr} hy_{gjlr}) \leq PY_{gil} + (1 - D_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{3.26}$$

$$PZ_{gil} + \sum_{r \in R} (r_{gr} HZ_{gilr} + q_{gr} WZ_{gilr} + p_{gr} LZ_{gilr}) \leq PZ_{gjl} + (1 - E_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{3.27}$$

$$PZ_{gjl} + \sum_{r \in R} (r_{gr} HZ_{gjlr} + q_{gr} WZ_{gjlr} + p_{gr} LZ_{gjlr}) \leq PZ_{gil} + (1 - F_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{3.28}$$

(**??**) ensures that if containers i and j are being used, they must be to the left, right, in front, behind, on top of or below one another.

$$A_{gijl} + B_{gijl} + C_{gijl} + D_{gijl} + E_{gijl} + F_{gijl} + (1 - X_{gil}) + (1 - X_{gjl}) \geq 1 \quad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{3.29}$$

(**??**) links the container type chosen with orientation.

$$LX_{gilr} + LY_{gilr} + LZ_{gilr} + WX_{gilr} + WY_{gilr} + WZ_{gilr} + HX_{gilr} + HY_{gilr} + HZ_{gilr} \leq 3 \cdot X_{gilr} \qquad \forall g \in G, r \in R, i \tag{3.30}$$

(**??**) - (**??**) restrict the solution so that containers are in the truck.

$$PX_{gil} + \sum_{r \in R} (p_{gr} LX_{gilr} + q_{gr} WX_{gilr} + r_{gr} HX_{gilr}) \leq p_k + (1 - X_{gil})\, M \quad \forall g \in G, i \in I, k \in K, t \in T \tag{3.31}$$

$$PY_{gil} + \sum_{r \in R} (q_{gr} WY_{gilr} + p_{gr} LY_{gilr} + r_{gr} ZY_{gilr}) \leq q_k + (1 - X_{gil})\, M \qquad \forall g \in G, i \in I, k \in K, t \in T \tag{3.32}$$

$$PZ_{gil} + \sum_{r \in R} (r_{gr} HZ_{gilr} + p_{gr} LZ_{gilr} + q_{gr} WZ_{gilr}) \leq r_k + (1 - X_{gil})\, M \qquad \forall g \in G, i \in I, k \in K, t \in T \tag{3.33}$$

A possible orientation of a container must satisfy that the length, width and height of container i are x-, y- or z-axis a single axis. Moreover, these have to be different from one

another. Therefore, the following constraints must hold.

$$LX_{gilr} + LY_{gilr} + LZ_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{3.34}$$
$$WX_{gilr} + WY_{gilr} + WZ_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{3.35}$$
$$HX_{gilr} + HY_{gilr} + HZ_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{3.36}$$
$$LX_{gilr} + WX_{gilr} + HX_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{3.37}$$
$$LY_{gilr} + WY_{gilr} + HY_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{3.38}$$
$$LZ_{gilr} + WZ_{gilr} + HZ_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{3.39}$$

To ensure trucks are loaded to minimum utilisation, the equation constrains the solutions so the truck must at least be loaded to α_k of its capacity.

$$\alpha_k \leq \frac{\sum_{c,l} V_{gklrt}\Gamma_c}{\Gamma_k} \qquad \forall g \in G, k \in K, t \in T \tag{3.40}$$

Finally, we define the domain of the variables.

$$I_{nt} \geq 0 \qquad \forall n \in N, t \in T \tag{3.41}$$
$$Y_{kt} \in \{0, 1\} \qquad \forall k \in K, t \in T \tag{3.42}$$
$$X_{gil}, X_{gilr}, X^n_{gilr} \in \{0, 1\} \qquad \forall r \in R, i \in I, k \in K, n \in N, t \in T \tag{3.43}$$
$$V^n_t, V^k_{int} \in \mathbb{N} \cup \{0\} \qquad \forall i \in I, k \in K, n \in N, t \in T \tag{3.44}$$
$$PX_{gil}, PY_{gil}, PY_{gil} \geq 0 \qquad \forall i \in I, t \in T, k \in K \tag{3.45}$$
$$LX_{gilr}, LY_{gilr}, LZ_{gilr}, WX_{gilr}, WY_{gilr}, WZ_{gilr}, HX_{gilr}, HY_{gilr}, HZ_{gilr} \in \{0, 1\} \qquad \forall r \in R, i \in I, k \in K, t \in T \tag{3.46}$$
$$A_{gijl}, B_{gijl}, C_{gijl}, D_{gijl}, E_{gijl}, F_{gijl} \in \{0, 1\} \qquad \forall i \in I, j \in J, k \in K, t \in T \tag{3.47}$$

Regarding vertical stability, the 3D-SCPTOP model is extended to ensure that loaded boxes resist the force of gravity. Similar to (**2016Paquay**), vertical stability is ensured by determining if the vertices of containers are supported by one another when they do not directly contact the ground. Furthermore, we impose that for an RTI to be in top of another, it has to have all four corners in top of another RTI, unless it's at the ground level. The four bottom corners are denoted by a tuple (u, u') where $u, u' \in U = \{0, 1\}$. Parameter u refers to the position on the x-axis (with $u = 0$ indicating a minimum x value and $u = 1$ a maximum x value). Parameter u' is defined analogously for the y-axis (see Figure **??**).

The following additional variables are defined:

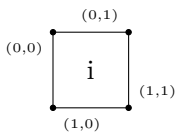


Figure 3.5: Description of the bottom corners position

Variables	
T_{gijl}	Binary variable equal to 1 if RTI $i \in I$ is directly on top of $j \in I$ in loading configuration l.
GL_{gil}	Binary variable indicating if container i is on the ground level on truck k during time period t.
$O_{gijuu'l}$	Binary variable equal to 1 if the bottom corner (u, u') of container i is immediately supported by container j on truck k during time period t.

Constraints (**??**) and (**??**) link the z-coordinate of container i with the ground–level variable GL_{gil}:

$$M\Big(1 - GL_{gil}\Big) \geq PZ_{gil} \quad \forall i \in I,\ t \in T \tag{3.48}$$

$$1 - GL_{gil} \leq M\, PZ_{gil} \quad \forall i \in I,\ t \in T \tag{3.49}$$

Constraints (**??**) and (**??**) imply that for $O_{gijuu'l}$ to be active, the heights of containers i and j must be compatible:

$$PZ_{gil} \leq PZ_{gjl} + \sum_{r \in R}\Big(r_{gr}\, HZ_{gjlr} + q_{gr}\, WZ_{gjlr} + p_{gr}\, LZ_{gjlr}\Big) + (1 - O_{gijuu'l})M \qquad \forall i, j \in I,\ i \neq j,\ k \in K,\ (u, u') \in U \times U,\ t \in T \tag{3.50}$$

$$PZ_{gil} + (1 - O_{gijuu'l})M \geq PZ_{gjl} + \sum_{r \in R}\Big(r_{gr}\, HZ_{gjlr} + q_{gr}\, WZ_{gjlr} + p_{gr}\, LZ_{gjlr}\Big) \qquad \forall i, j \in I,\ i \neq j,\ k \in K,\ (u, u') \in U \times U,\ t \in T \tag{3.51}$$

Furthermore, the coordinates of the corner (u, u') of container i must lie within the XY projection of container j. Constraints (**??**) and (**??**) link the x-dimensions, and constraints (**??**) and (**??**) link the y-dimensions. Parameter ϵ assures that the corner (u, u') has

at least ϵ support.

$$PX_{gil} + u\left(\sum_{r\in R}(p_{gr}\,LX_{gilr} + q_{gr}\,WX_{gilr} + r_{gr}\,HX_{gilr})\right) + M(1 - O_{gijuu'l}) \geq PX_{gjl} + \epsilon\,u \quad \forall i,j \in I,\ i \neq j,\ k \in K,\ (u,u') \in U \times U,\ t \in T \tag{3.52}$$

$$PX_{gil} + u\left(\sum_{r\in R}(p_{gr}\,LX_{gilr} + q_{gr}\,WX_{gilr} + r_{gr}\,HX_{gilr})\right) \leq PX_{gjl} + \sum_{r\in R}(p_{gr}\,LX_{gjlr} + q_{gr}\,WX_{gjlr} + r_{gr}\,HX_{gjlr}) + M(1 - O_{gijuu'l}) - (1-u)\epsilon \quad \forall i,j \in I,\ i \neq j,\ k \in K,\ (u,u') \in U \times U,\ t \in T \tag{3.53}$$

$$PY_{gil} + u'\left(\sum_{r\in R}(q_{gr}\,WY_{gilr} + p_{gr}\,LY_{gilr} + r_{gr}\,HY_{gilr})\right) + M(1 - O_{gijuu'l}) \geq PY_{gjl} + \epsilon\,u' \quad \forall i,j \in I,\ i \neq j,\ k \in K,\ (u,u') \in U \times U,\ t \in T \tag{3.54}$$

$$PY_{gil} + u'\left(\sum_{r\in R}(q_{gr}\,WY_{gilr} + p_{gr}\,LY_{gilr} + r_{gr}\,HY_{gilr})\right) \leq PY_{gjl} + \sum_{r\in R}(q_{gr}\,WY_{gjlr} + p_{gr}\,LY_{gjlr} + r_{gr}\,HY_{gjlr}) + M(1 - O_{gijuu'l}) - (1-u')\epsilon \quad \forall i,j \in I,\ i \neq j,\ k \in K,\ (u,u') \in U \times U,\ t \in T \tag{3.55}$$

Constraint (**??**) ensures that either a used container is at the bottom of the truck or all four bottom corners are supported by other boxes:

$$\sum_{j\in I,\ (u,u')\in U\times U} O_{gijuu'l} + M\Big(GL_{gil} + 1 - X_{gil}\Big) \geq 4 \quad \forall i \in I,\ k \in K,\ t \in T \tag{3.56}$$

Constraint (**??**) limits $O_{gijuu'l}$ to be active only if container j is used during planning period t:

$$O_{gijuu'l} \geq X_{gjl} \quad \forall g \in G,\ i,j \in I,\ (u,u') \in U \times U,\ k \in K,\ t \in T \tag{3.57}$$

Additionally, to allow only 90 degree rotations along the z-axis the following constraint is added:

$$HZ_{gilr} \leq X_{gilr} \quad \forall g \in G,\ r \in R,\ i \in I,\ k \in K,\ t \in T \tag{3.58}$$

Finally, the following equations link the RTI stacking variables:

$$4\,T_{gijl} = \sum_{(u,u')\in U\times U} O_{gijuu'l} \quad \forall g \in G,\ i,j \in I,\ l \in L,\ i \neq j$$

$$\sum_{j\in I} T_{gijl} \leq 1 \quad \forall g \in G,\ i \in I,\ l \in L$$

3.5 Matheuristic Procedure

As previously mentioned, exact methods provided by MILP solvers fall short for tackling real world instances with the current computation technology. Therefore, a matheuristic approach is proposed, Math-3D-SCPTOP. It is based on decoupling the two sub-processes involved in the problem: finding 3D loading configurations of RTIs inside the trucks and determining the number of trucks and what to carry in each time period in order to minimize inventory and transport costs.

This matheuristic is based on two main components. First a MILP model is defined to determine which loading configurations from a given set should be used in order to minimise the transport and inventory costs. Second a metaheuristic approach is used to identify these loading configurations.

3.5.1 PLC-SCPTOP

The following MILP model, dubbed as PLC-SCPTOP, uses predefined loading configurations of container types to determine which loading configurations to use, the products to carry and the number of trucks in order to minimise inventory and transport costs. The predefined loading configurations must already obey the spatial constraints defined in 3D-SCPTOP. Theoretically, if every conceivable loading configuration were incorporated into the model, the optimal solution of PLC-SCPTOP would align with that of 3D-SCPTOP. Enumerating all of the possible loading configurations of a 3D Bin Packing Problems would take an exponential number of memory space and cpu time with respect the number of container types. Moreover, solving the PLC-SCPTOP would become even more time-consuming, assuming the MILP program could be fully allocated in memory.

Table **??**

Table 3.5: Nomenclature of Math-3D-SCTOP

Indices	
R	Set of RTI types ($r = 1, \ldots, R$)
K	Set of trucks ($k = 1, 2, \ldots, K$)
N	Set of products *(n = 1,. . . ,N)*
L	Set of loading configurations ($l = 1, 2, \ldots, L$)
T	Set of planning periods *(t =1, 2. . . T)*
G	Set corresponding to the supplier and plant nodes $G = \{s, p\}$
Set of indices	
$M(r)$	Set of RM or AP that are carried in RTI $r \in R$, $n = 1, \ldots, M(r)$
Parameters	
cK	Cost of using a truck
lt	Lead transport time between the plant and the supplier and viceversa.
i_{gn}	Initial inventory level of $n \in N$ in $g \in G$ in terms of full RTIs
ie_{gr}	Initial inventory level of empty RTIs $r \in R$ in $g \in G$ RTIs
ic_{gn}	Inventory cost of $n \in N$ at $g \in G$ in terms of full RTIs
d_{gnt}	Production plan of $n \in N$ at $g \in G$ during time period $t \in T$
q_{rt}	Initial quantity of empty RTIs $r \in R$ arriving at supplier $s \in G$ in time period $t \in (1, \ldots, lt)$
q_{nt}	Initial quantity of full RTIs of $n \in R$ arriving at p in time period $t \in (1, \ldots, lt)$

lc_{lrg}	Number of containers of type c used in loading configuration l arriving at $g \in G$
ss_{gnt}	Security stock for product $n \in N$ in time period $t \in T$ in $g \in G$ in terms of full RTIs
ss_{gtr}	Security stock of empty RTIs $r \in R$ in time period $t \in T$ in $g \in G$
w_n	Weight of a container carrying product n
k_t	Numbers of trucks available at time period t
pK	Maximum weight capacity of a truck
Decision Variables	
I_{gnt}	Inventory of product $n \in N$ at the end of $t \in T$ time period at $g \in G$ in terms of full RTIs
Ie_{gtr}	Inventory of empty RTIs $r \in R$ at the end of $t \in T$ time period at $g \in G$
Y_{gkt}	Binary variable, if one then truck $k \in K$ is used in time period $t \in T$ arriving at $g \in G$ and zero otherwise
Y_{gklt}	Binary variable, if one then truck $k \in K$ is used in time period $t \in T$ departing from $g \in G$ using loading configuration l, 0 otherwise
Q_{nt}	Integer variable indicating the number of RM or AP $n \in N$ in transport during time period t to the manufacturing plant p
Q_{rt}	Integer variable the number of empty RTIs $r \in R$ in transport during time period t to the supplier s
V_{gklrt}	Number of RTIs $r \in R$ in truck $k \in K$ using loading configuration $l \in L$ at time period $t \in T$ departing from $g \in G$
V_{kglnt}	Number of full RTIs with RM or AP $n \in N$ in truck $k \in K$ using loading configuration $l \in L$ at time period $t \in T$ departing from $g \in G$

We wish to minimise the total cost, which is given by transportation and inventory costs.

$$Min\ z = \sum_{g \in G} \sum_{k \in K} \sum_{t \in T} Y_{gkt} \cdot cK + \sum_{g \in G} \sum_{n \in N} \sum_{t \in T} I_{gnt} \cdot ic_{gn} \tag{3.59}$$

(**??**) - (**??**) define the inventory level of RM or AP according to the production demand and supply of RM or AP from the supplier s at the plant p

$$I_{pnt} = I_{pnt'} - d_{pnt} + Q_{nt'} \qquad \forall n \in N, t \in T : t \geq lt + 1, t' := t - lt \tag{3.60}$$

$$I_{pnt} = I_{pnt'} - d_{pnt} + q_{nt} \qquad \forall n \in N, t \in (2, \ldots, lt) :, t' := t - lt \tag{3.61}$$

$$I_{pnt} = i_{pn} - d_{pnt} + q_{nt} \qquad \forall n \in N : t := 1 \tag{3.62}$$

$$Ie_{ptr} = Ie_{pt'r} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \qquad \forall r \in R, t \in T : t \geq 2 \tag{3.63}$$

$$Ie_{ptr} = ie_{pr} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \qquad \forall r \in R, t := 1 \tag{3.64}$$

(**??**) - (**??**) the empty RTIs according to the production demand and supply of RTIs from

the plant p at the supplier s

$$I_{snt} = I_{snt'} + d_{snt} - Q_{nt} \qquad \forall n \in N, t \in T, t' \in T, : t \geq 2, t' := t - lt \tag{3.65}$$

$$I_{snt} = i_{sn} + d_{snt} - Q_{nt} \qquad \forall r \in R : t := 1 \tag{3.66}$$

$$Ie_{str} = Ie_{st'r} + Q_{rt'} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R, t \in T : t \geq lt + 1, t' := t - lt \tag{3.67}$$

$$Ie_{str} = Ie_{st'r} + q_{rt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R, t \in (2, \ldots, lt) :, t' := t - lt \tag{3.68}$$

$$Ie_{str} = ie_{pr} + q_{rt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R : t = 1 \tag{3.69}$$

Each product must satisfy a minimum security stock ss_{gnt}.

$$I_{gnt} \geq ss_{gnt} \quad \forall g \in G, n \in N, t \in T \tag{3.70}$$

(**??**) imposes that a truck k can only use one loading configuration l with direction $g \in G$ in time period t.

$$\sum_{l \in L} Y_{gklt} = Y_{gkt} \quad \forall k \in K, g \in G, t \in T \tag{3.71}$$

(**??**) - (**??**) impose that the truck k at time period t using loading configuration l uses the same loading configuration as l.

$$V_{gklrt} + M \cdot (1 - Y_{gklt}) \geq lc_{lrg} \qquad \forall k \in K, g \in G, l \in L, t \in T \tag{3.72}$$

$$V_{gklrt} \leq Y_{gklt} \cdot lc_{lrg} + M \cdot (1 - Y_{gklt}) \qquad \forall k \in K, l \in L, g \in G, t \in T \tag{3.73}$$

Since a product can only be carried by one container type (**??**) sets that the number of containers of type c carried at truck k using loading configuration l at time period c is equal to the sum of containers carried of products n which containers of type c can carry. Moreover, (**??**) sets the total amount of product n being transported from s to p in time period t, and (**??**) sets the total amount of RTIs r being transported from p to s in time period t,

$$V_{pklrt} = \sum_{n \in M(r)} V_{pklnt} \qquad \forall k \in K, l \in L, t \in T, r \in R \tag{3.74}$$

$$Q_{nt} = \sum_{l \in L} \sum_{k \in K} V_{kslnt} \qquad \forall n \in N, t \in T \tag{3.75}$$

$$Q_{rt} = \sum_{l \in L} \sum_{k \in K} V_{pklrt} \qquad \forall r \in R, t \in T \tag{3.76}$$

(**??**) guarantees that the maximum weight allowed for truck k is not surpassed.

$$\sum_{l,n} w_n V_{kglnt} \leq pK \qquad \forall k \in K, t \in T, g \in G \tag{3.77}$$

$$
\begin{aligned}
I_{gnt} &\in \mathbb{Z}_{\geq 0} && \forall g \in G,\ n \in N,\ t \in T, \\
Ie_{gtr} &\in \mathbb{Z}_{\geq 0} && \forall g \in G,\ t \in T,\ r \in R, \\
Y_{gkt} &\in \{0,1\} && \forall k \in K,\ g \in G,\ t \in T, \\
Y_{gklt} &\in \{0,1\} && \forall k \in K,\ g \in G,\ l \in L,\ t \in T, \\
Q_{nt} &\in \mathbb{Z}_{\geq 0} && \forall n \in N,\ t \in T, \\
Q_{rt} &\in \mathbb{Z}_{\geq 0} && \forall r \in R,\ t \in T, \\
V_{gklrt} &\in \mathbb{Z}_{\geq 0} && \forall g \in G,\ k \in K,\ l \in L,\ r \in R,\ t \in T, \\
V_{kglnt} &\in \mathbb{Z}_{\geq 0} && \forall k \in K,\ g \in G,\ l \in L,\ n \in N,\ t \in T.
\end{aligned}
$$

3.5.2 Meta-Heuristic Procedure

The objective of the metaheuristic is to obtain good loading configurations so the PLC-SCPTOP model can optimise transport and inventory costs while managing RTI flow. Therefore, the metaheuristic should consider the temporal nature of the problem. That is, the loading problem must consider the future demand of products and the current state of inventories.

To simulate the production and transportation, we define two separate processes. On the one hand, we have the plant processes and on the other we have the supplier processes. In each node, we first receive the RTIs, then we produce, and after that we make the transport decisions. **??** illustrates the both processes and the connections between them.

In both cases, we need to determine transport decisions. That is, the loading configurations and what is going to be transported. We can only transport what we already have in stock. Furhtermore, RTIs are loaded into the truck as stacks. Each RTI has a set of compatible RTIs which can be stacked with. We use a column generation technique to find the RTIs stacks that can be loaded. Then, we find a loading configuration through a 2D-SLOPP formulation, where the stacks that contain RTIs needed in the other supply chain node for the next products are prioritized through the objective function. Once the RTI loading configuration is determined, we decide RM or AP procurement for the manufacturing site, prioritizing RMs or APs which have less stock at the manufacturing site.

We will use the previously found loading configurations to consider the time dependency of the loading problem at hand. For each period $t \in T$, we will solve the PLC-SCPTOP problem from day 1 to t with the current loading configurations obtained to find the inventory levels of the products at day t. Then, we find the necessary containers for the following $\delta \in \Delta$ days, where Δ is a user-defined list of parameters.

Finally, we feed the loading configurations found by Algorithms **??** and **??** to the PLC-SCPTOP mathematical modelling as parameters to find reasonable quality solutions to 3D-SCPTOP problem.

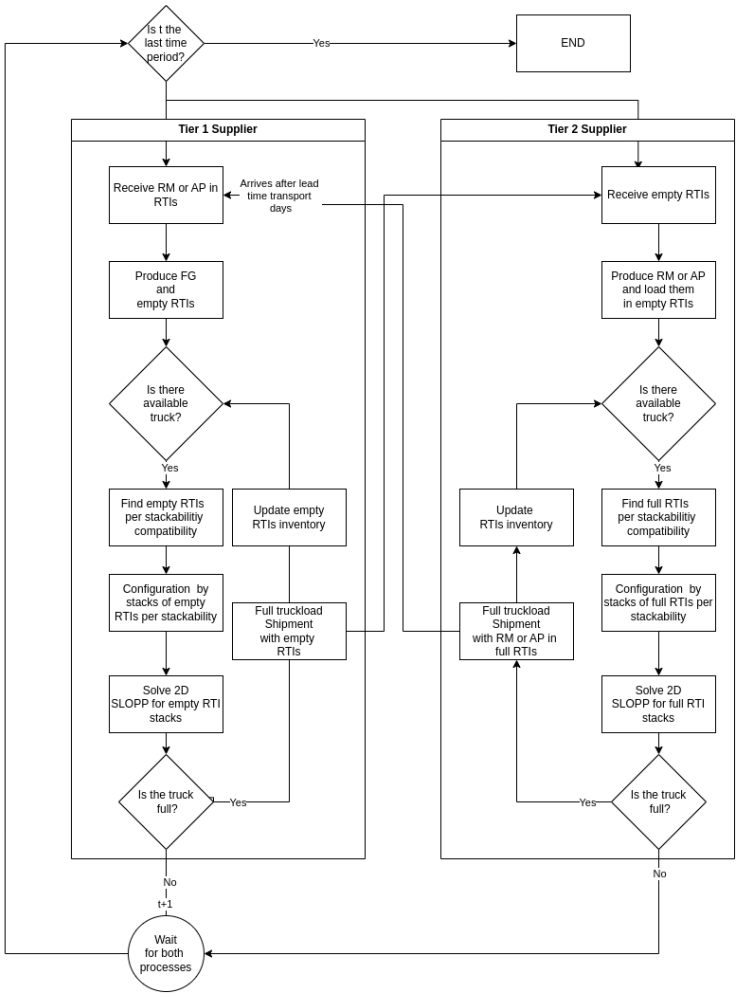


Figure 3.6: Procceses of plant and supplier

Loading Procedure

To address the loading problem, will divide the problem in two parts. First, we will find through a column generation technique for the cutting stock problem to find different stacks that will be loaded. Once these stacks are found, we will

Algorithm 1 Loading Configurations

```
procedure LoadingConfigurations
    T # List of time periods
    LC=[] # List of loading configurations
    for t ∈ T do
        for δ ∈ Δ do
            IncomingTransport(t)
            Produce(t)
            while TransportResourceAvailable do
                C = ContainersToLoad(t, δ, LC)


                LC.append(PackContainers(C,t))
            end while
        end for
    end for
    LC=EliminateDuplicateOrDominatedLoadingConfigurations(LC)
    return LC
end procedure
```

Algorithm 2 Final Loading Configurations

```
procedure Final Loading Configurations(LC)
    TimePeriods # List of time spans to calculate the number of needed prod-
ucts
    TimeSpan # List of time spans to calculate the number of needed products
    for t in TimePeriods do
        for δ in TimeSpan do
            I=SolvePLC-SCPTOP(LC,t)
            C= ContainersToLoad(I,t, δ, LC)
            LC.append(PackContainers(C,t))
        end for
    end for
    LC=EliminateDuplicateOrDominatedLoadingConfigurations(LC)
    return LC
end procedure
```

3.6 Computational and comparative analysis

Here the 3D-SCPTOP model is validated and compared to the work by (**Diaz-Madronero2014APlanning**). This fuzzy optimisation model integrates procurement transport and inventory holding decisions by contemplating transport capacity restrictions by a maximum number of containers

Algorithm 3 MathH-3D-SCPTOP

1: Δ_1, Δ_2 # List of time spans to calculate the number of needed products
2: LC = InitialLoadingConfiguration(Δ_1)
3: LC = FinalLoadingConfiguration(LC, Δ_2)
4: SolvePLCSCPTOP(LC,|T|))

per truck, which we refer to as SCPTOP from this point onwards. To compare the SCPTOP model, we make three key modifications: fuzzy parameters are not considered; transport capacity restriction takes into account the volume of individual containers and the truck's capacity; products can be loaded in more than one container type. The SCPTOP model is extended and defined in Appendix **??**. We evaluate and compare both the 3D-SCPTOP and SCPTOP models, in terms of solution quality, transport costs, inventory costs and computational times. The 3D-SCPTOP and SCPTOP mathematical formulations are solved for all the instance sets with CP-SAT from OR-Tools and Gurobi using their default settings. Additionally, as ($C\&P$) problems are NP-hard, the Meta-3D-SCPTOP algorithm has been developed to find igh-quality solutions for large real-world instances, based on a case study. So the performance of the 3D-SCPTOP, the Meta-3D-SCPTOP and the SCPTOP models is compared and evaluated as follows.

Twenty-four instances are artificially generated by a stochastic process, whose key characteristics are given as follows:

- **Demand Generation**: the demand for each product is randomly generated by deviating from a given mean by plus or minus ten percent. This introduces variability into the demand values for different products
- **Time Periods**: Half of the instances have a total of five time periods, while the other half have a total of ten time periods. This allows evaluating the model's performance over different time horizons.
- **Container Types**: Each instance includes a varying number container types, specifically two, three or four. The distribution of container types is the same among instances, which implies that an equal number of instances has the same number of containers
- **Container Dimensions**: The dimensions of each container type are randomly generated. The width, length and height of boxes are fractions of the width, length, or height of trucks. This randomness ensures diversity in the sizes of containers across the generated instances. An additional constraint is applied to the generation of containers. The volume of each container must be at least 1/n of the truck's volume, where n can take the values of four, six or eight. This ensures that containers are appropriately sized in relation to trucks

By incorporating these stochastic elements into the generation process, the 24 instances provide a diverse set of scenarios for evaluating the model's performance. Variability in demand, time periods, box types and box dimensions introduces realistic and challenging conditions that the model needs to address. Furthermore, we use the number of time periods, the number of box types, the number of products and the maximum amount of containers needed to load a truck (in volume terms) as explanatory variables of various metrics from the models, such as total cost, transport cost, inventory cost, solution gap and solution time.

The instance name is in the form of $I_n_t_c_v$, where n is product type, t is the number of time periods, c is the number of container types and v is the minimum volume of containers.

Each test is limited to 4 hours (14,400 seconds) of run-time per instance. Tests are performed on a Microsoft Windows Server 2022 Standard n with an AMD EPYC 7402 24-Core Processor running at 2,800 MHz, with 256 GB of RAM using Python as the programming language.

3.7 Practical application

This section presents a practical procurement transport application. The proposed resolution framework was evaluated using real data from an automotive industry supply chain. We validate and compare the 3D-SCPTOP solution procedures to the company's procedure, which employs predefined loading configurations that indicate the number of each container type to be carried. Finding these loading configurations is a very tedious and time-consuming manual task that is based on a spreadsheet procedure. A mathematical programming model, dubbed as PLC-SCPTOP, which replicates the current's company procedure, is then utilised to minimise the total cost, with a summation of procurement transport and inventory holding costs, and one based on these predetermined configurations.

The real-world data derived from the company's operations act as the empirical foundation for examining both the PLC-SCPTOP and the Meta-3D-SCPTOP. To broaden the scope of this analysis, we scale and manipulate these data, guided by a stochastic process, to simulate several potential operational scenarios. Setting our analysis to mirror the dynamic and unpredictable realities of supply chain operations thus enhances the validity of our evaluation and the practical relevance of the suggested metaheuristic. For this purpose, we propose three different scenarios for the created instances: normal demand, low variability demand, high variability demand. Table **??**, found in the Appendix **??**outlines the models' performance across these instances. Table **??** summarises models' performance and resolution times.

Table 3.6: Summary of performance for the practical application

Model	Objective Value	Best Bound	Duration (seconds)
MathH-3D-SCTOP	252235.198934	247307.820849	740.000000
Meta-3D-SCTOP	256715.145777	NaN	677.555556
PLC-SCTOP	275226.680415	271798.477640	44.670370

Figures **??** - **??** describe the performance of both the models and solution procedures in the different scenarios. The x-axis shows the scenario, while the y-axis depicts the objective value, inventory costs, or transport costs. Each colour represents a different model applied to obtain the solution.

The results reveal that an effective systematic method is proposed for solving procurement transport operational planning with 3D full-truck loading constraints. This solution is both automated and executed within a reasonable computational timeframe. Our methodology proves particularly beneficial when there is significant discrepancy between the company's proposed loading configurations and the actual demand concerning container types. In such instances, the company is usually left with two choices: either use the fixed loading configurations and, thus, incur higher inventory and transport costs, or manually resolve the

loading configuration via a spreadsheet procedure, which might lead to errors. However, one potential disadvantage of the proposed Meta-3D-SCPTOP is its performance under normal demand conditions because the demand for different container types is consistent with the loading configurations, but the metaheuristic may result in fewer optimal inventory holding costs compared to the company's current procedure. This is because the Meta-3D-SCPTOP obtains decisions locally, while exact methods are guaranteed, given enough time, to find the optimal solution.

This issue is, however, solved when utilising MathH-3D-SCPTOP because it effectively combines both heuristic and exact methods. While the heuristic provides a fast and reasonably good solution for the initial loading configurations, the exact method improves it by further lowering inventory and transport costs.

Therefore, this method shows promising results for reducing inventory and transport costs, particularly under variable demand and misalignment between loading configurations and container types.

3.8 Conclusions

We herein propose a MILP model, 3D-SCPTOP, to tackle procurement transportation planning at an operational level by incorporating 3D loading constraints into its considerations. Given the NP-hard nature of 3D bin packing, we have also developed a metaheuristic approach, Meta-3D-SCPTOP, which is later refined and enhanced by employing a mathheuristic technique, MatH-3D-SCPTOP.

A case study has also been conducted within an automotive SC that comprises a car assembler, a first-tier supplier and a second-tier supplier. The objective of the 3D-SCPTOP model is twofold: (i) to minimise the total number of trucks used; (ii) to minimise inventory holding costs. This approach takes into account the complexities of supply chain logistics in the automotive industry by demonstrating a successful integration of procurement transport planning and 3D loading constraints.

Managerial implications in a SC context are oriented to the proposed solution methodology, improving the solutions found by the company's procedure. It is also able to adapt to demand changes while finding high-quality solutions in reasonable computational times.

Nevertheless, several challenges remain for further research lines to deal with; for instance, the inclusion of multiple suppliers to better represent complex supply chains. A more comprehensive transport operation model could be achieved by incorporating routing aspects, such as vehicle routing with 3D loading constraints. Furthermore, the introduction of dynamic stability into loading constraints may also yield more practical, robust solutions. Integrating sourcing and delivery transport decisions could improve supply chain efficiency. Finally, the incorporation of production scheduling into the model has the potential of offering a cohesive approach to supply chain planning optimisation by aligning production timelines with transportation and inventory strategies.

.1 Mathematical Model

Table 7: Nomenclature of Supply Chain Operation Transport Planning

Sets of indices	
I:	Set of products ($i = 1, 2, \ldots, I$)
J :	Set of groups composed of products that must be ordered together ($j = 1, 2, \ldots, J$)
K :	Set of trucks ($k = 1, 2, \ldots, K$)
T :	Set of planning periods (days) ($t = 1, 2, \ldots, T$)
Decision variables	
$Q_{ikt:}$	Units transported of i by k in period t (units)
G_{ijkt} :	Units transported of i corresponding to group j by k in period t (units)
C_{kt} :	Volume of containers transported by k in period t (units)
I_{it} :	Inventory amount of i at the end of period t (units)
$K_{jkt:}$	Number of lots to order of products of group j by k in period t
Y_{kt}	Binary variable indicating whether a truck k has been used in period t
Objective functions	
z_1 :	Total number of trucks utilized
z_2	Total inventory amount generated
Parameters	
u_i :	Amount of product i that fits in a container (units)
l_j :	Number of units of each group lot j (units)
b_{ij}	1 if product i belongs to group j, and 0 otherwise
D_{it} :	Demand of product i in t (units) (considered firm)
Γ_k :	Volume of truck k
Γ_i :	Volume of container carrying product i
m :	Minimum truck occupation (in containers)
$I0_i$	Inventory amount of i in period 0
IC_i	Inventory cost of i
C_k	Cost of using truck k
SC_i	Number of days of stock coverage for product i

Minimize the total cost transport and inventory

$$\text{Min}\, z \cong \sum_{k=1}^{K}\sum_{t=1}^{T} Y_{kt} + \sum_{i=1}^{I}\sum_{t=1}^{T} I_{it} \cdot IC_i$$

subject to

$$\begin{aligned}
&I_{it} = I_{i(t-1)} - D_{it} + \sum_{k=1}^{K} Q_{ikt} \quad \forall i \in I, t \in T,\\
&Q_{ikt} = \sum_{j=1}^{J} G_{ijkt} \quad \forall i \in I, k \in K, t \in T,\\
&G_{ijkt} = K_{jkt} \cdot l_j \cdot b_{ij} \quad \forall i \in I, j \in I, k \in K, t \in T,\\
&C_{kt} = \sum_{i=1}^{l} Q_{ikt}/u_i \cdot \Gamma_i \quad \forall k \in K, t \in T,\\
&C_{kt} \leqslant \Gamma_k \cdot Y_{kt} \quad \forall k \in K, t \in T,\\
&C_{kt} \geqslant m \cdot \Gamma_k Y_{kt} \quad \forall k \in K, \in T\\
&I_{nt} \geq \begin{cases} \sum_{s\in\{1,\ldots,SC_i\}} D_{i,t+s} & \text{if } i \in I, t \in \{1,\ldots,|T| - SC_i\}\\ \sum_{s\in\{1,\ldots,SC_i\}} D_{i,|T|-SC_i+s} & \text{other wise}\end{cases}\\
&Y_{kt} \leqslant 1 \quad \forall k, t,\\
&I_{it}, Q_{ikt}, G_{ijkt}, C_{kt}, K_{jkt}, Y_{kt} \geqslant 0 \text{ integer} \quad \forall i \in I, j \in I, k \in K, t \in T.
\end{aligned}$$

In our case $|J| = |I|$, one group for each product.